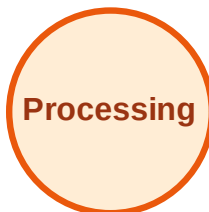


BFS Traversal

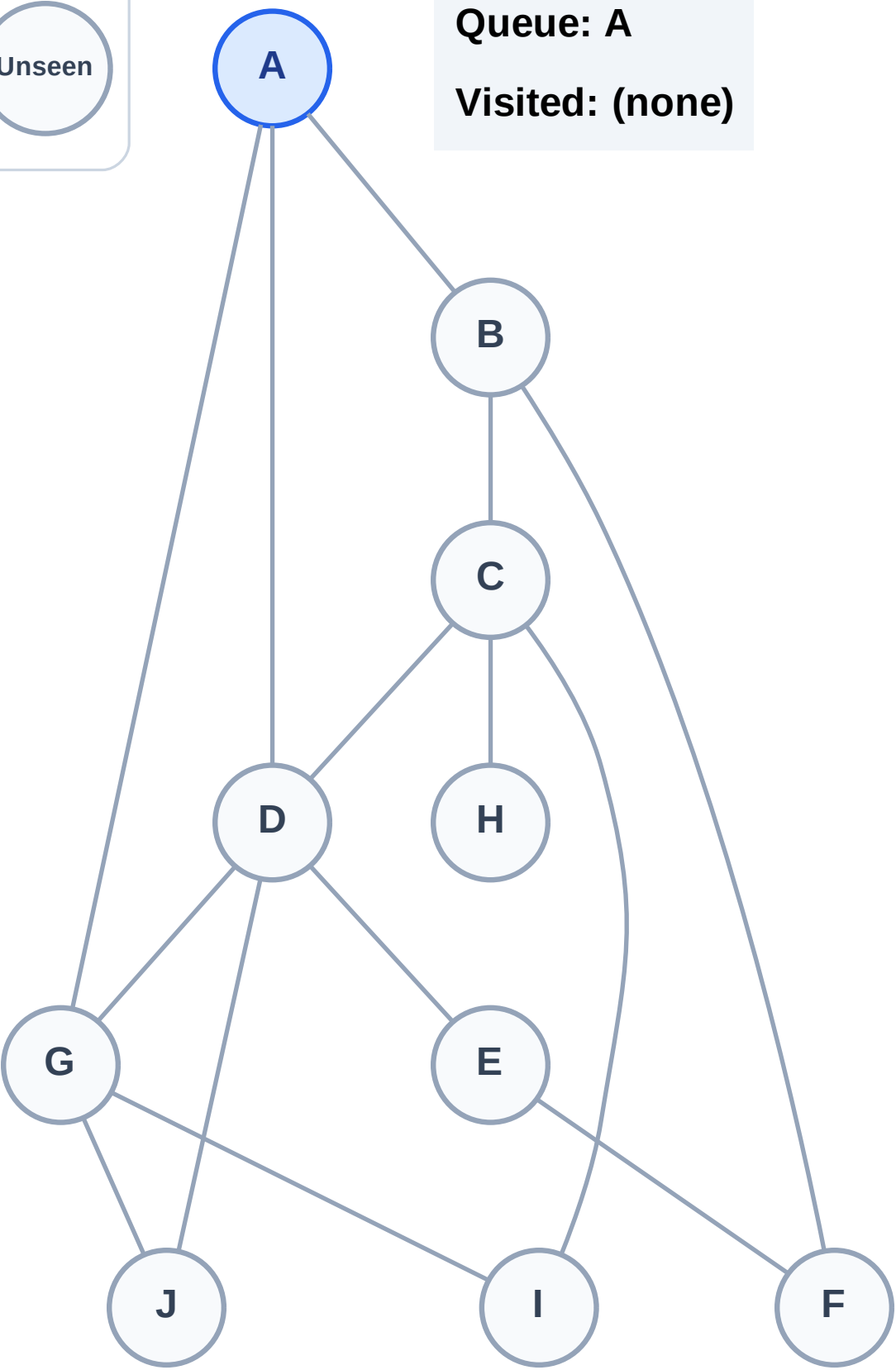
Step 1: Start at A

Legend



Queue: A

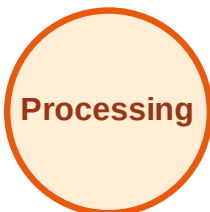
Visited: (none)



BFS Traversal

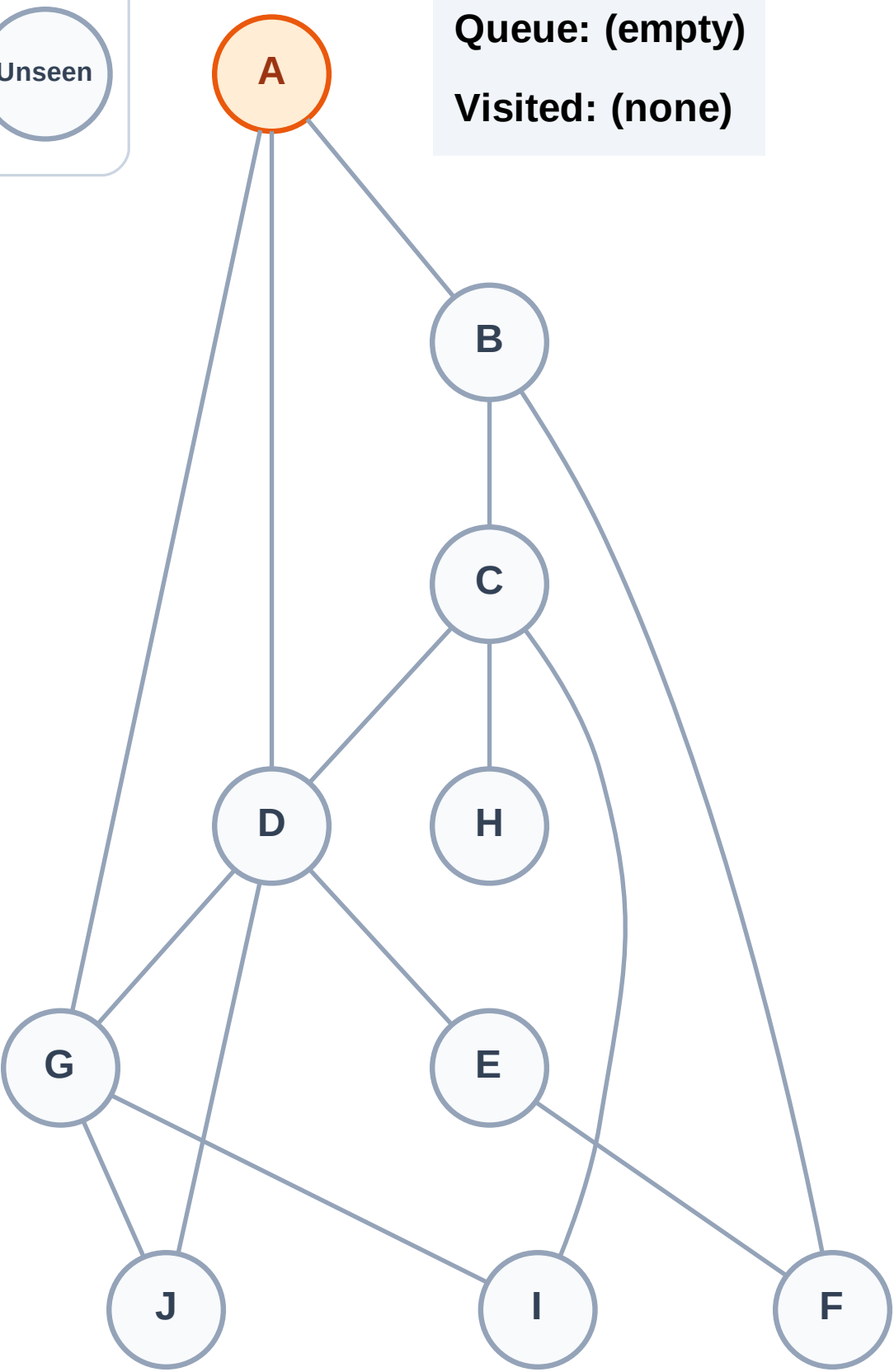
Step 2: Process node A

Legend



Queue: (empty)

Visited: (none)



BFS Traversal

Step 3: Discover B, D, G from A

Legend

Complete

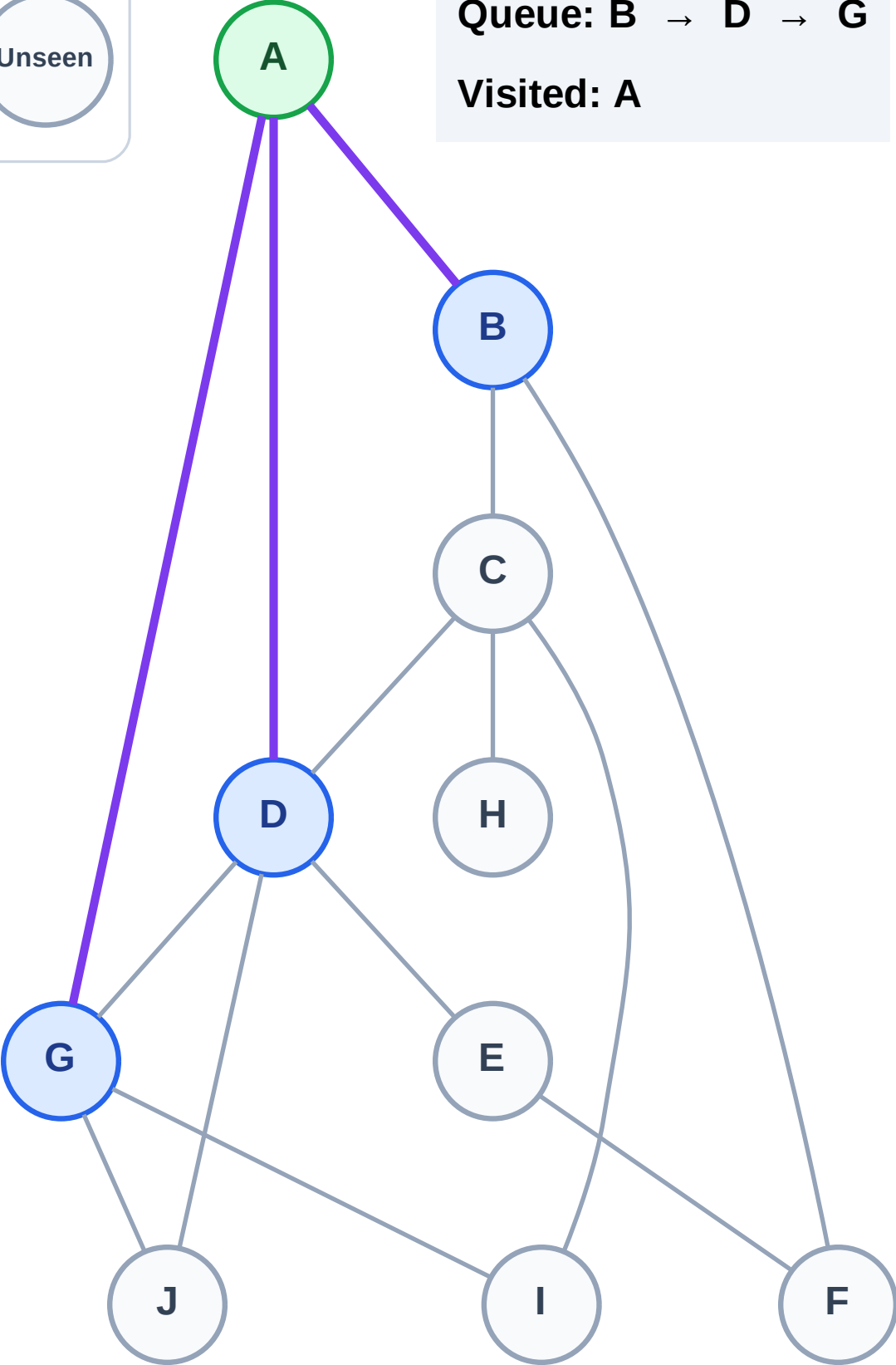
Processing

Discovered

Unseen

Queue: B → D → G

Visited: A



BFS Traversal

Step 4: Process node B

Legend

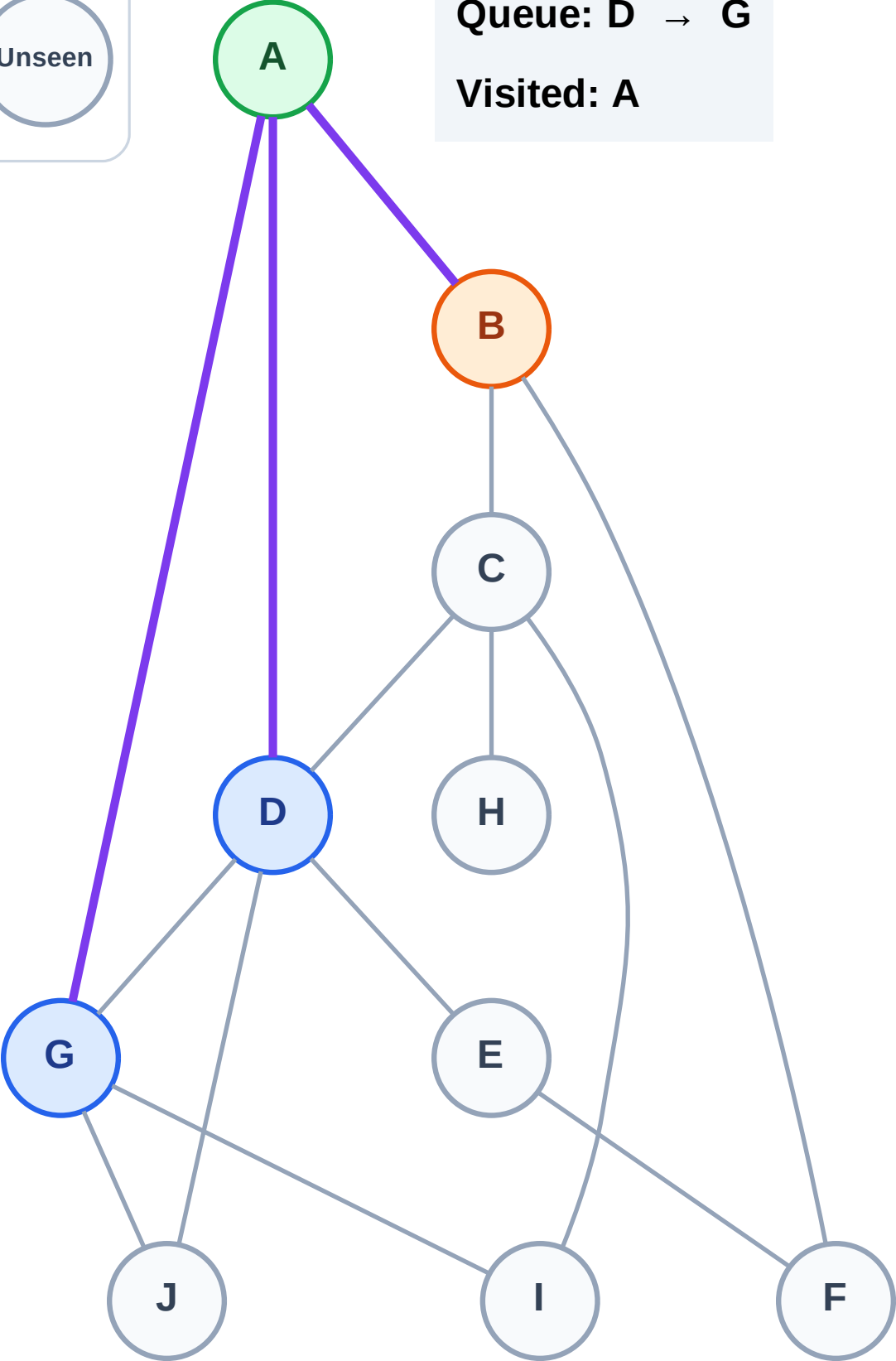
Complete

Processing

Discovered

Unseen

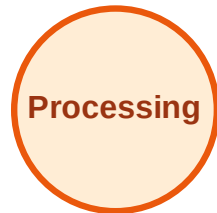
Queue: D → G
Visited: A



BFS Traversal

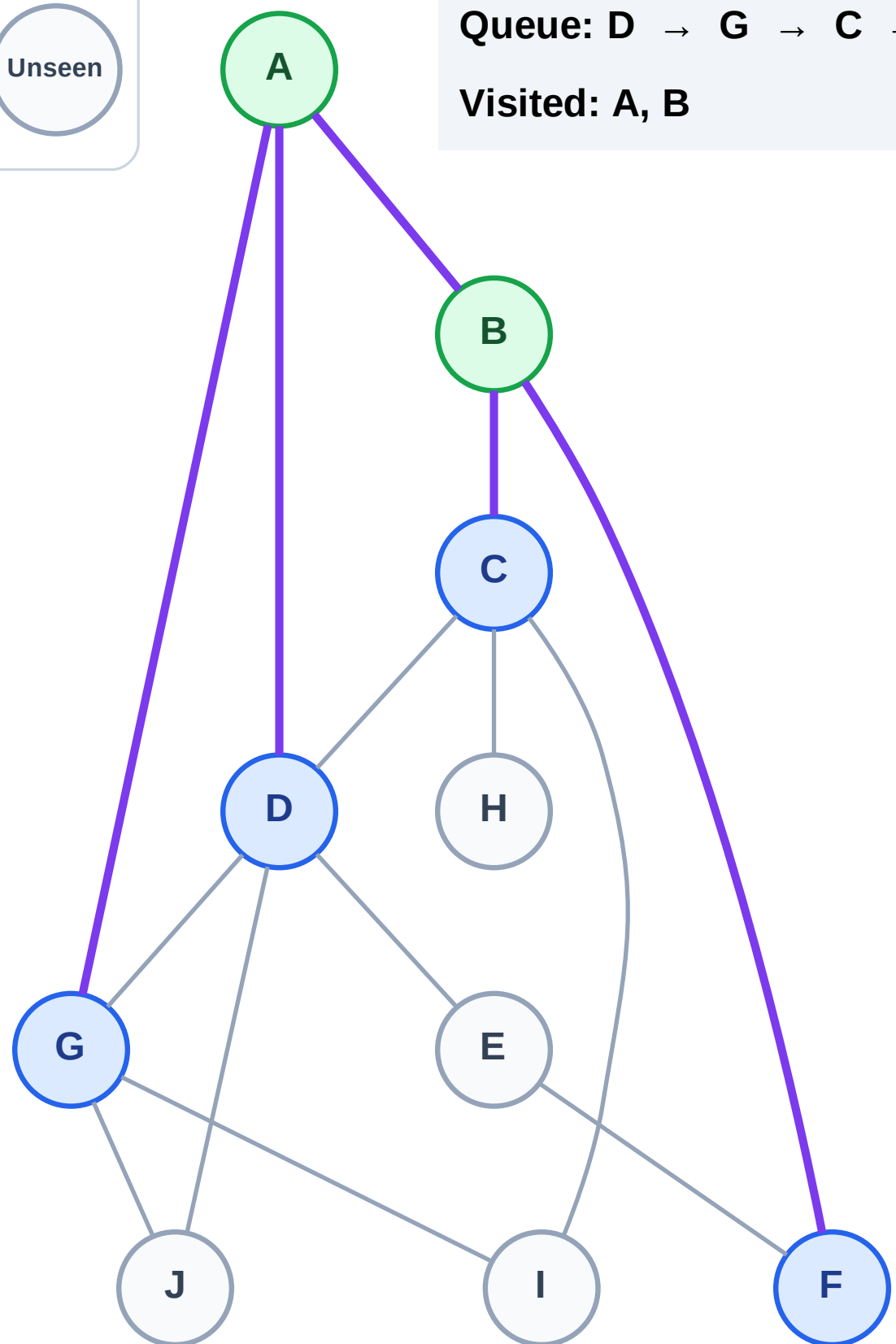
Step 5: Discover C, F from B

Legend



Queue: D → G → C → F

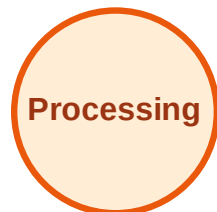
Visited: A, B



BFS Traversal

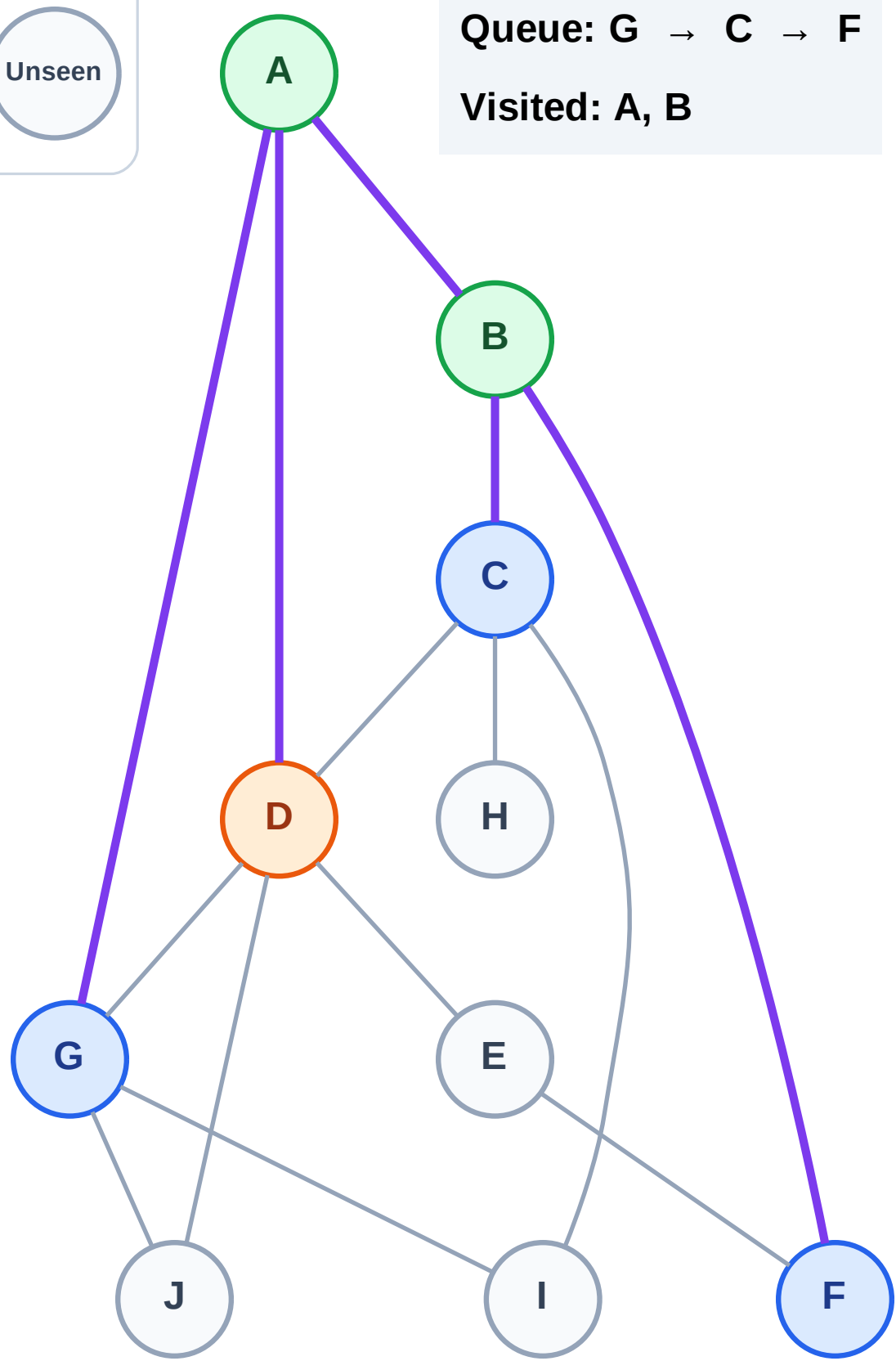
Step 6: Process node D

Legend



Queue: G → C → F

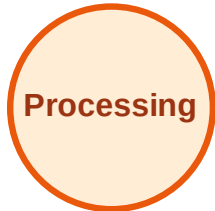
Visited: A, B



BFS Traversal

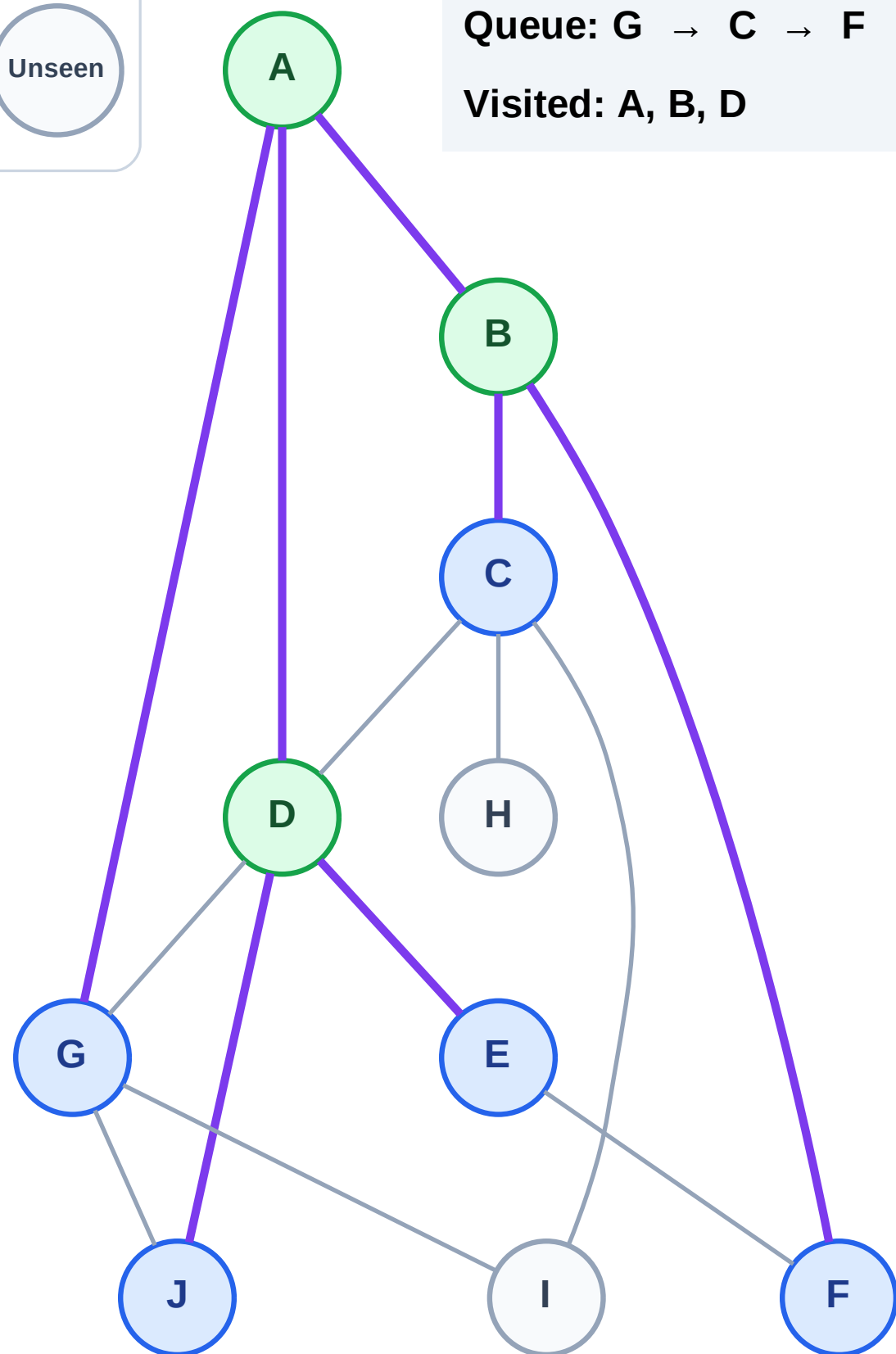
Step 7: Discover E, J from D

Legend



Queue: G → C → F → E → J

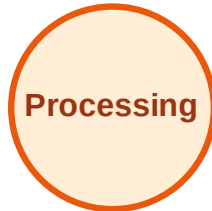
Visited: A, B, D



BFS Traversal

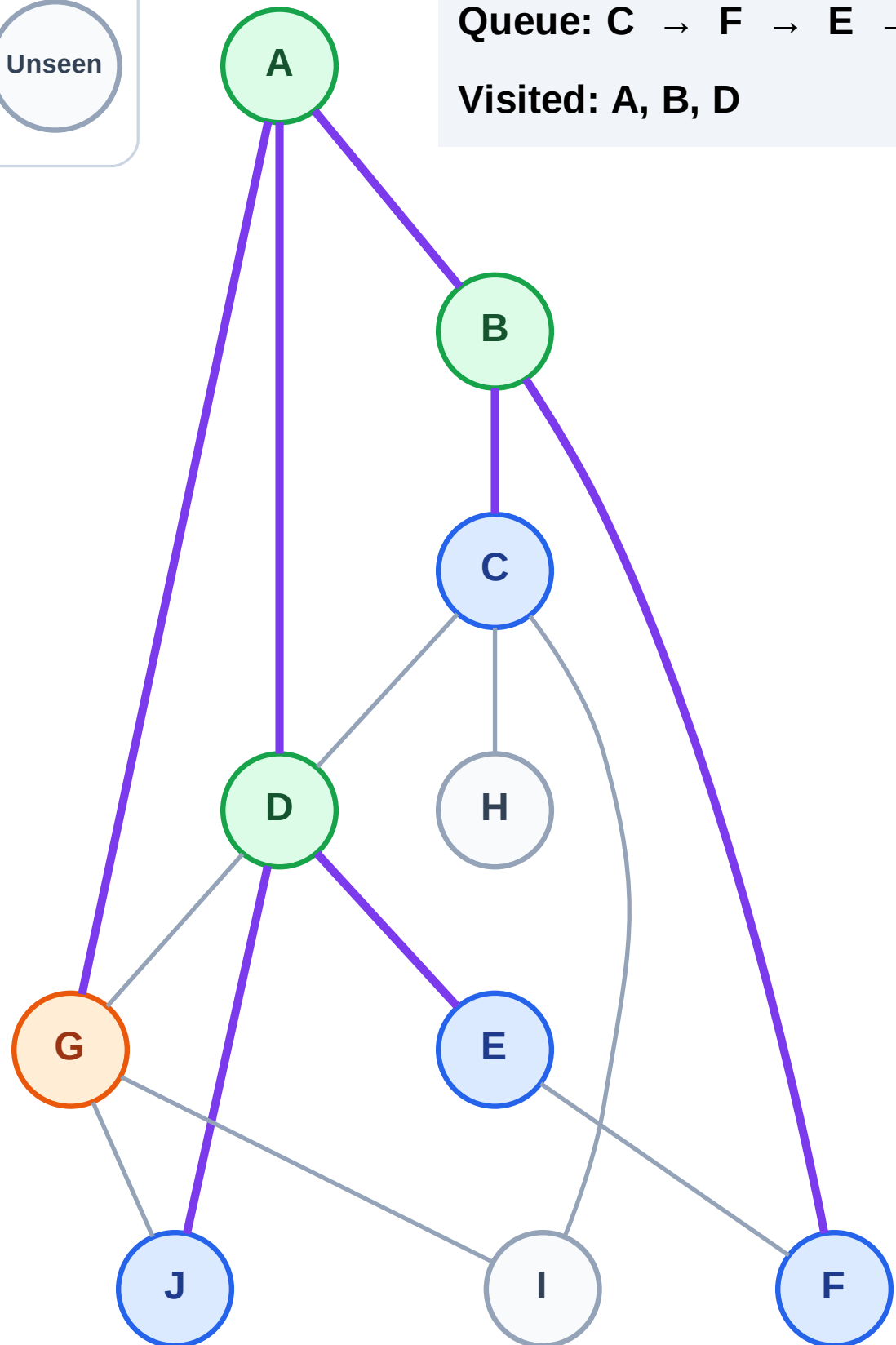
Step 8: Process node G

Legend



Queue: C → F → E → J

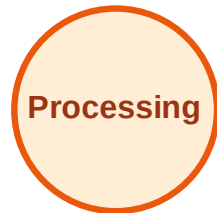
Visited: A, B, D



BFS Traversal

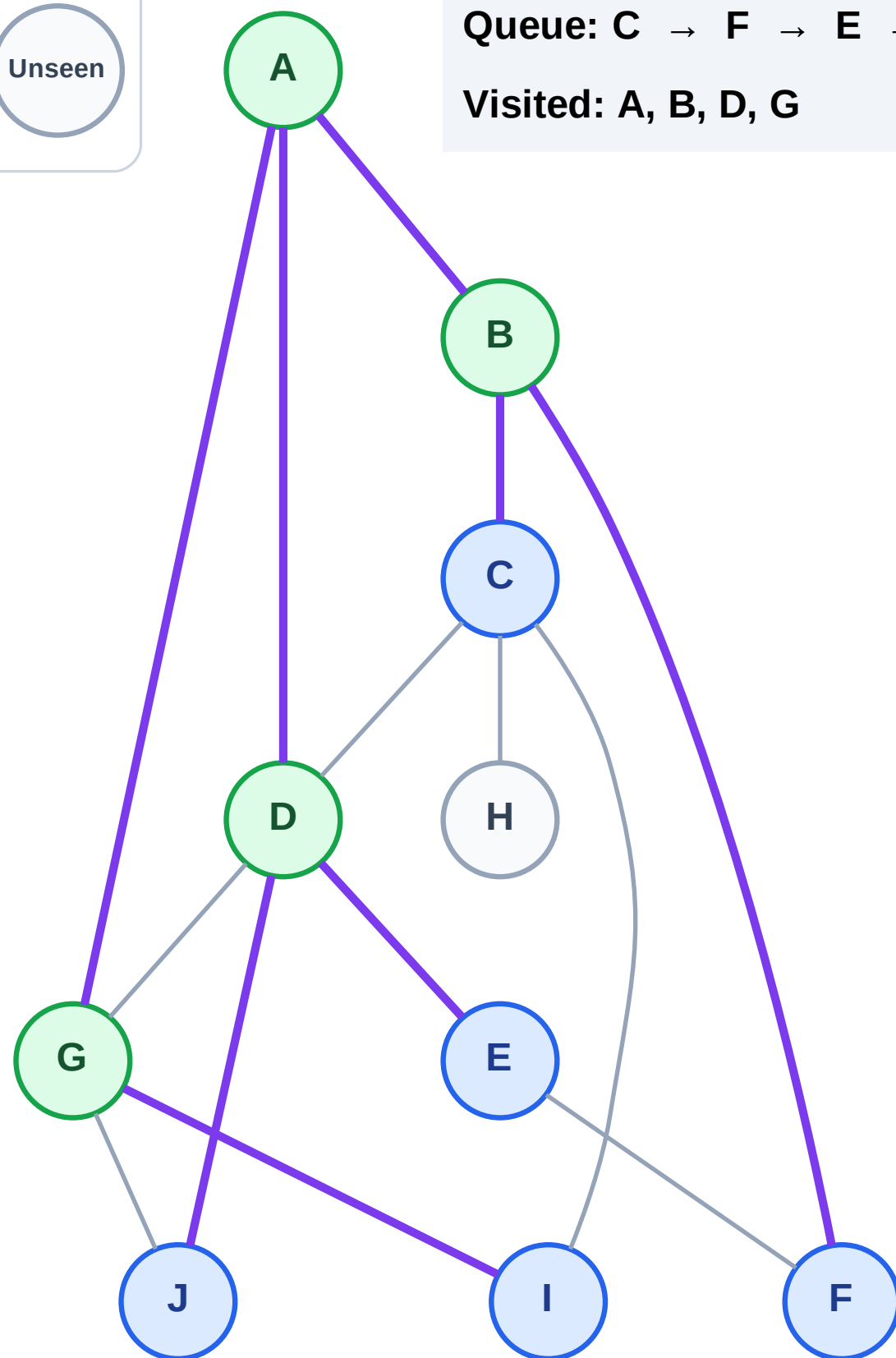
Step 9: Discover I from G

Legend



Queue: C → F → E → J → I

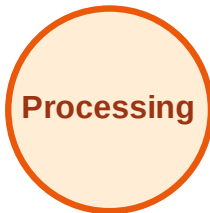
Visited: A, B, D, G



BFS Traversal

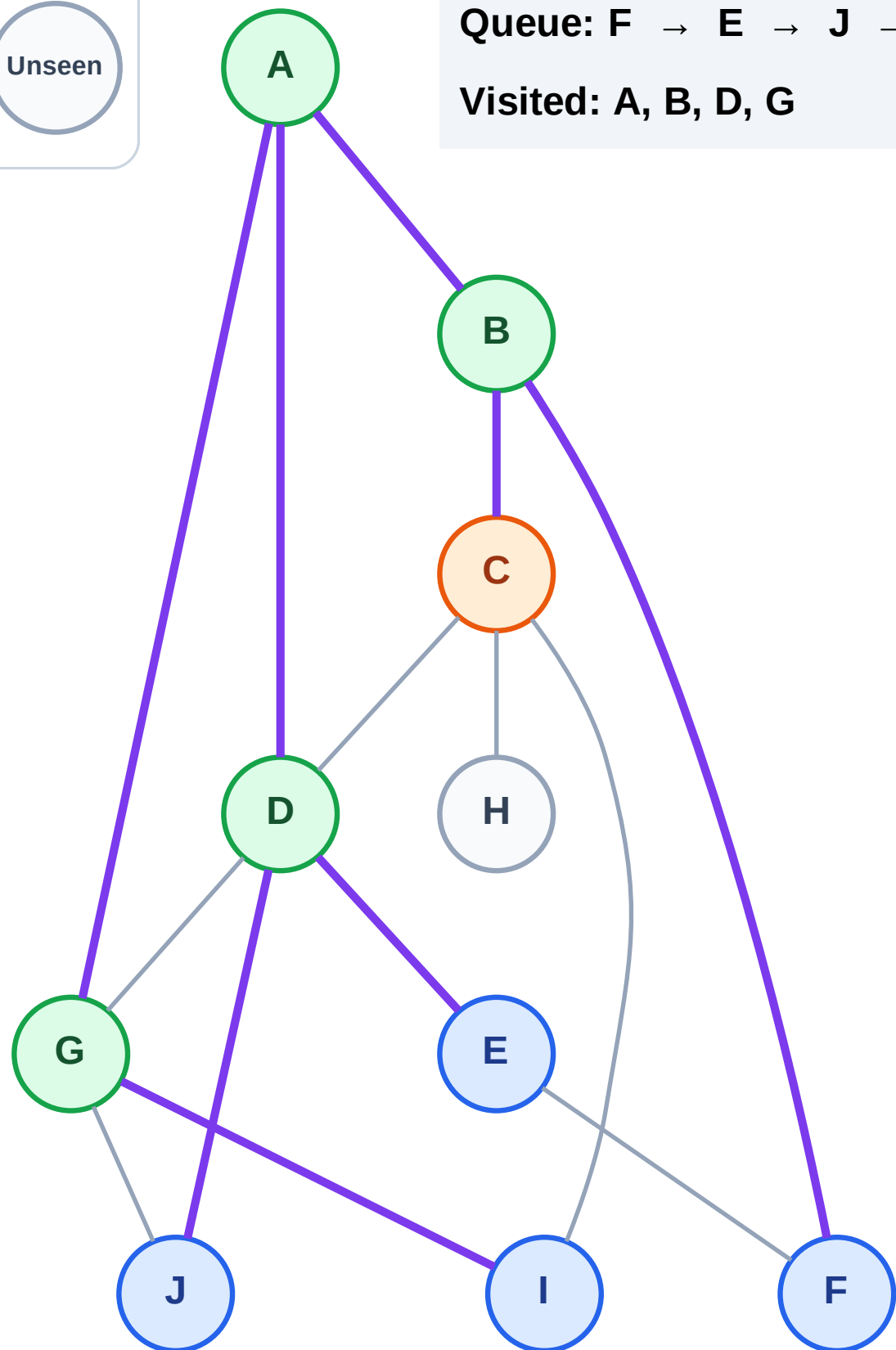
Step 10: Process node C

Legend



Queue: F → E → J → I

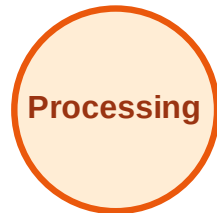
Visited: A, B, D, G



BFS Traversal

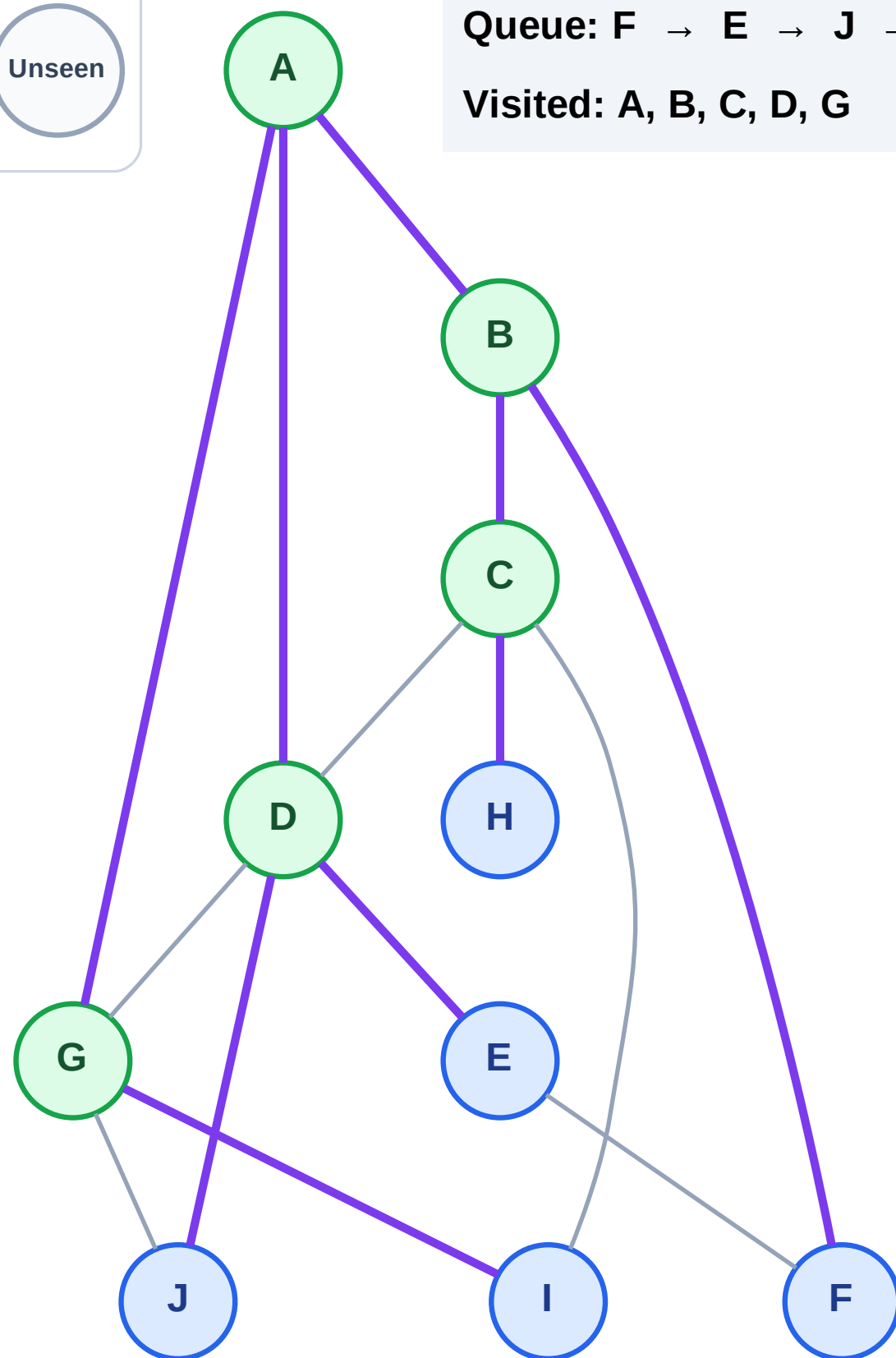
Step 11: Discover H from C

Legend



Queue: F → E → J → I → H

Visited: A, B, C, D, G



BFS Traversal

Step 12: Process node F

Legend

Complete

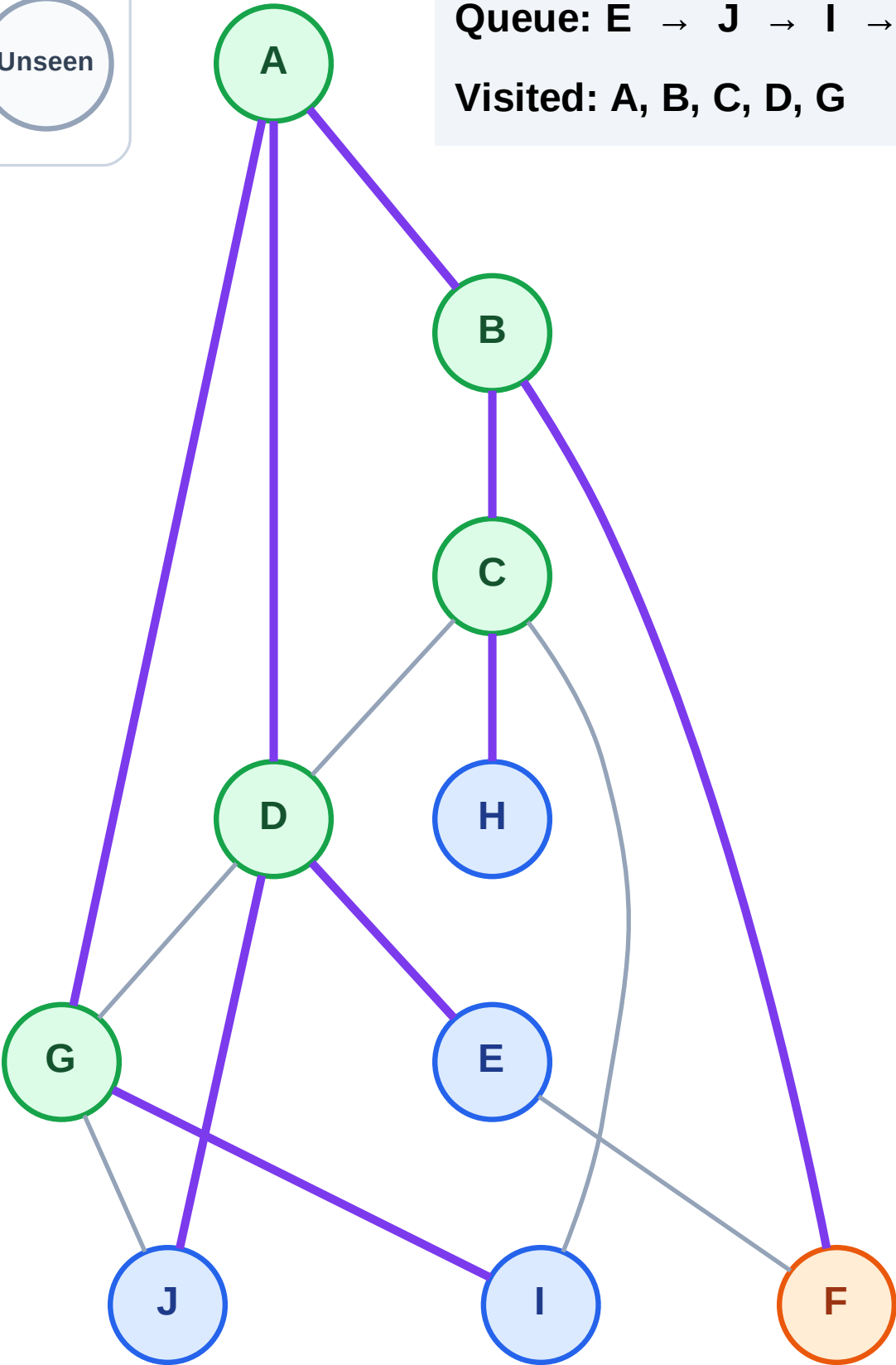
Processing

Discovered

Unseen

Queue: E → J → I → H

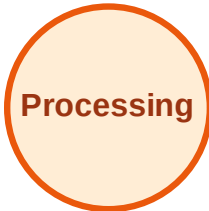
Visited: A, B, C, D, G



BFS Traversal

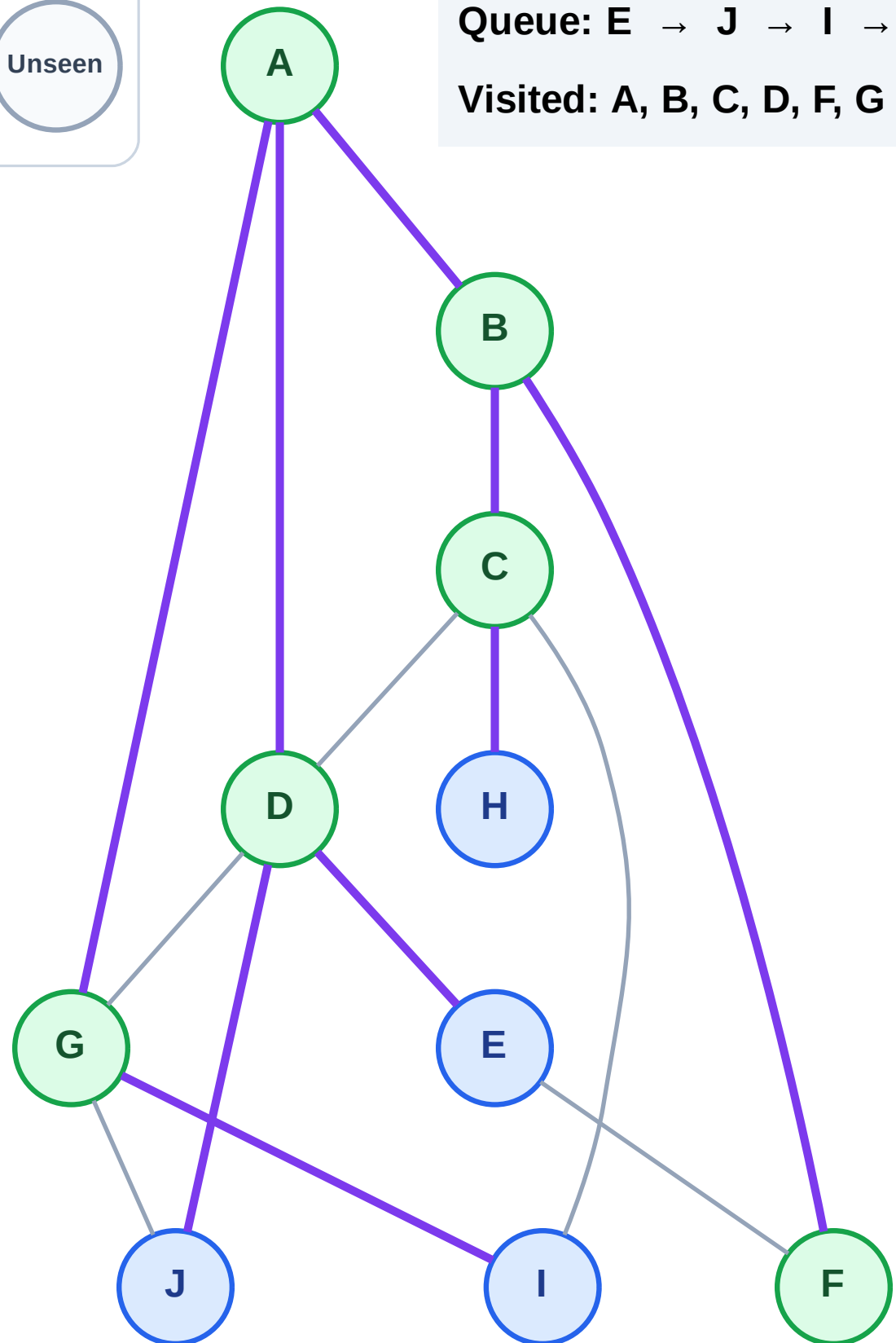
Step 13: No new neighbors from F

Legend



Queue: E → J → I → H

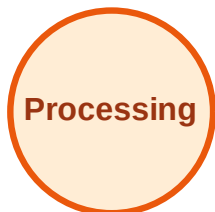
Visited: A, B, C, D, F, G



BFS Traversal

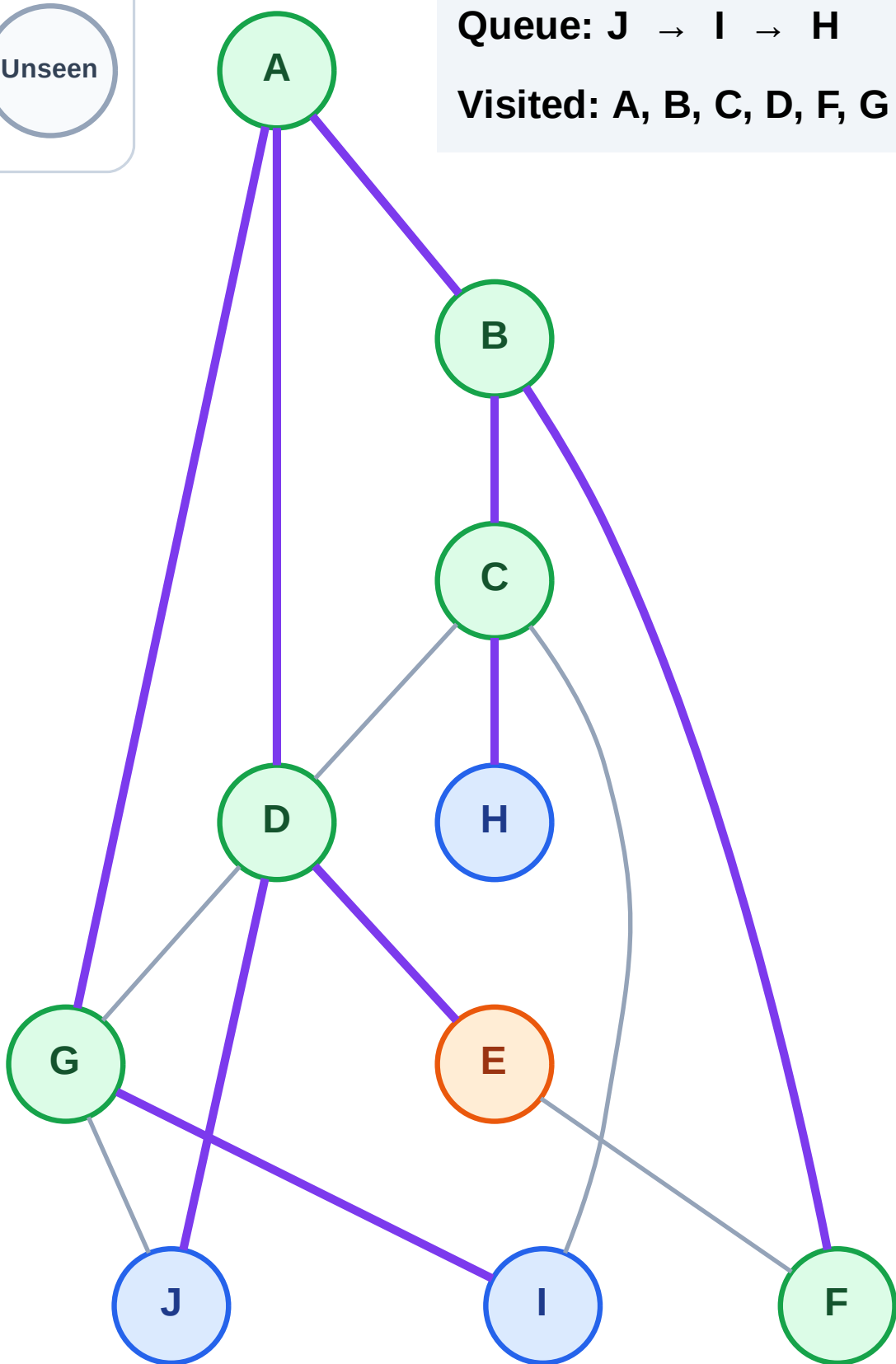
Step 14: Process node E

Legend



Queue: J → I → H

Visited: A, B, C, D, F, G



BFS Traversal

Step 15: No new neighbors from E

Legend

Complete

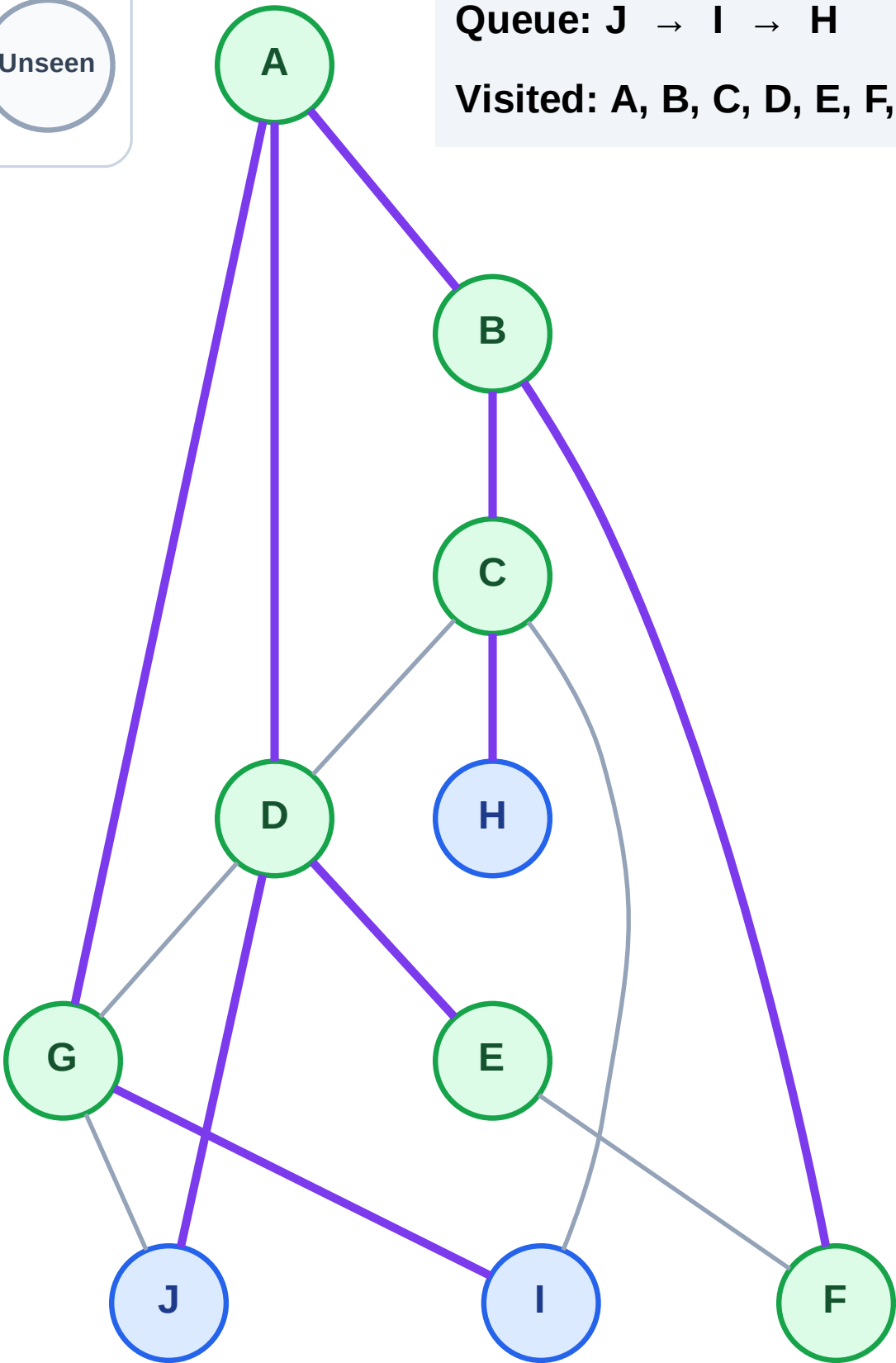
Processing

Discovered

Unseen

Queue: J → I → H

Visited: A, B, C, D, E, F, G



BFS Traversal

Step 16: Process node J

Legend

Complete

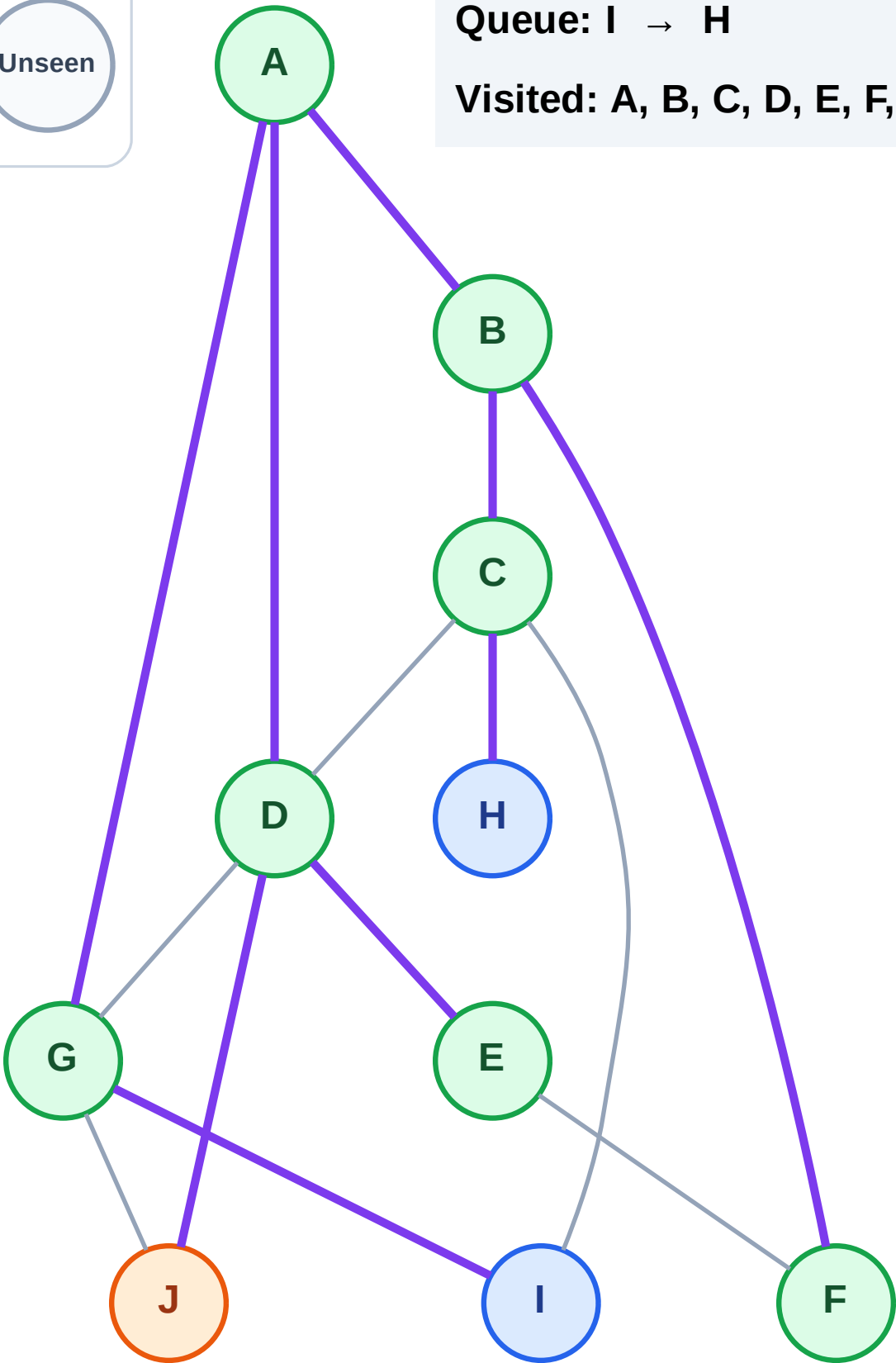
Processing

Discovered

Unseen

Queue: I → H

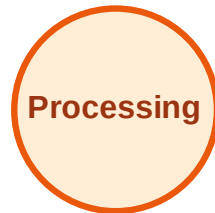
Visited: A, B, C, D, E, F, G



BFS Traversal

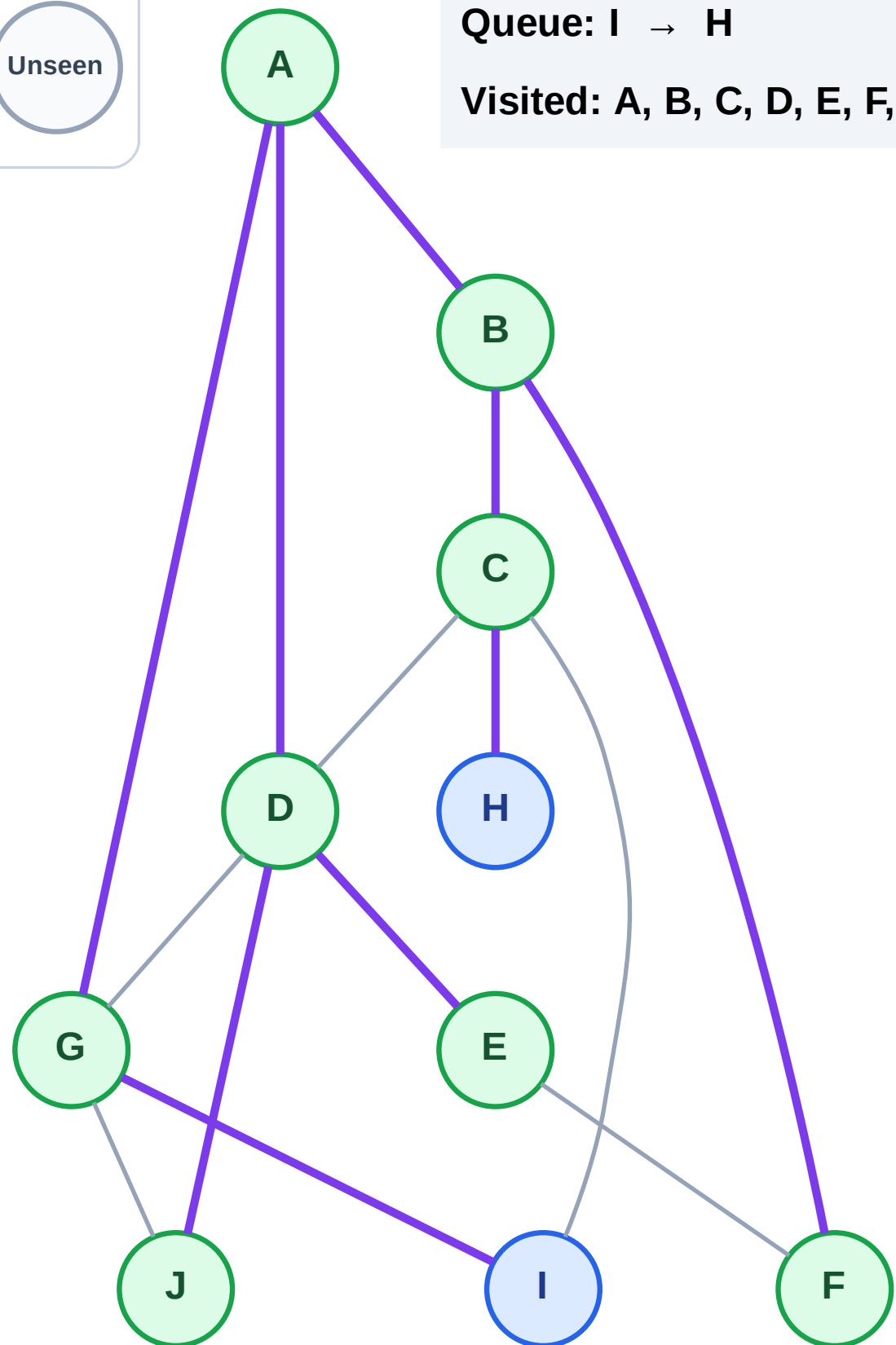
Step 17: No new neighbors from J

Legend



Queue: I → H

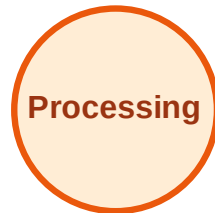
Visited: A, B, C, D, E, F, G, J



BFS Traversal

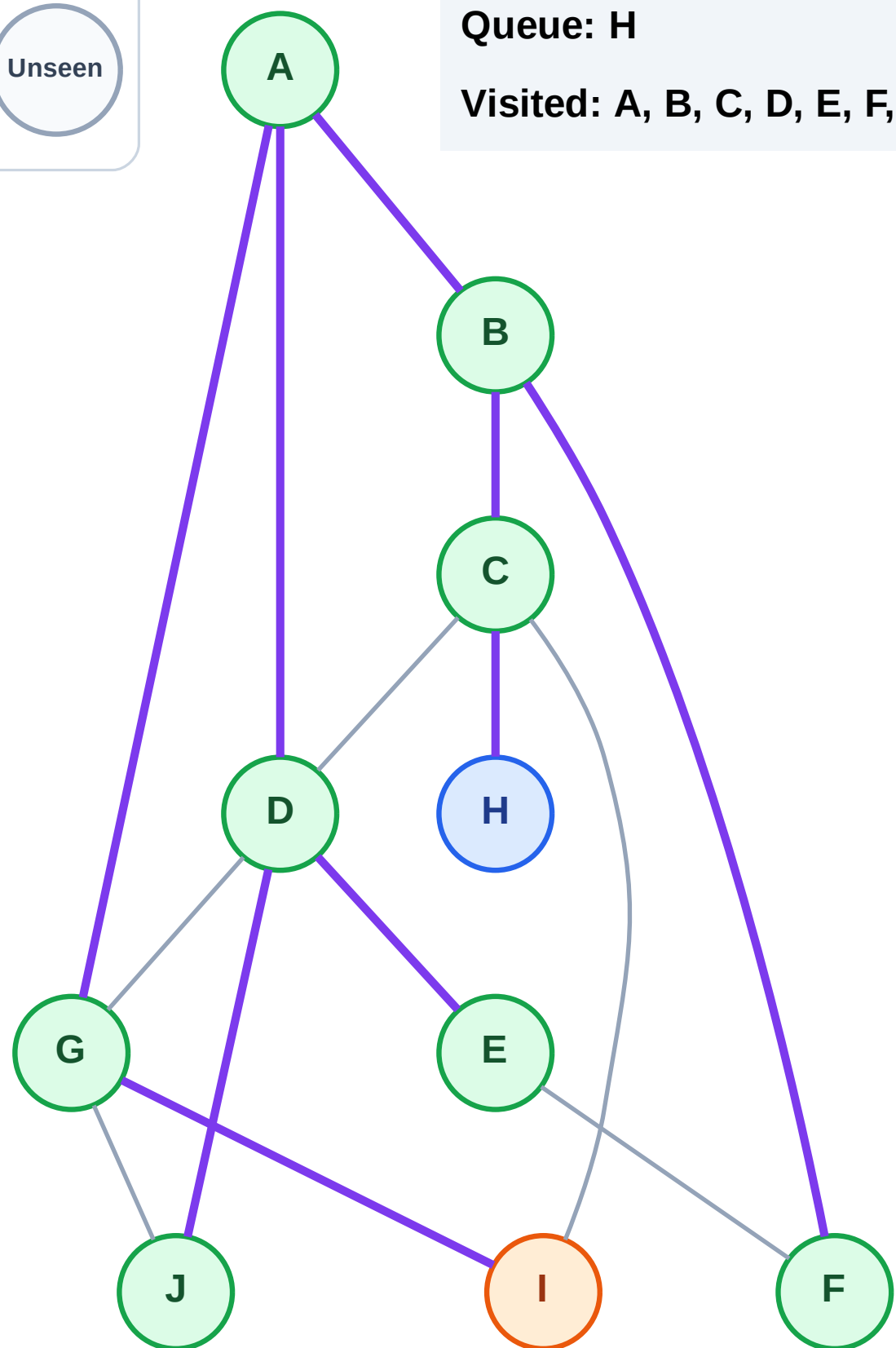
Step 18: Process node I

Legend



Queue: H

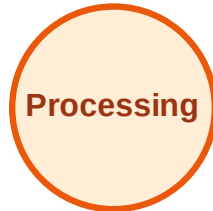
Visited: A, B, C, D, E, F, G, J



BFS Traversal

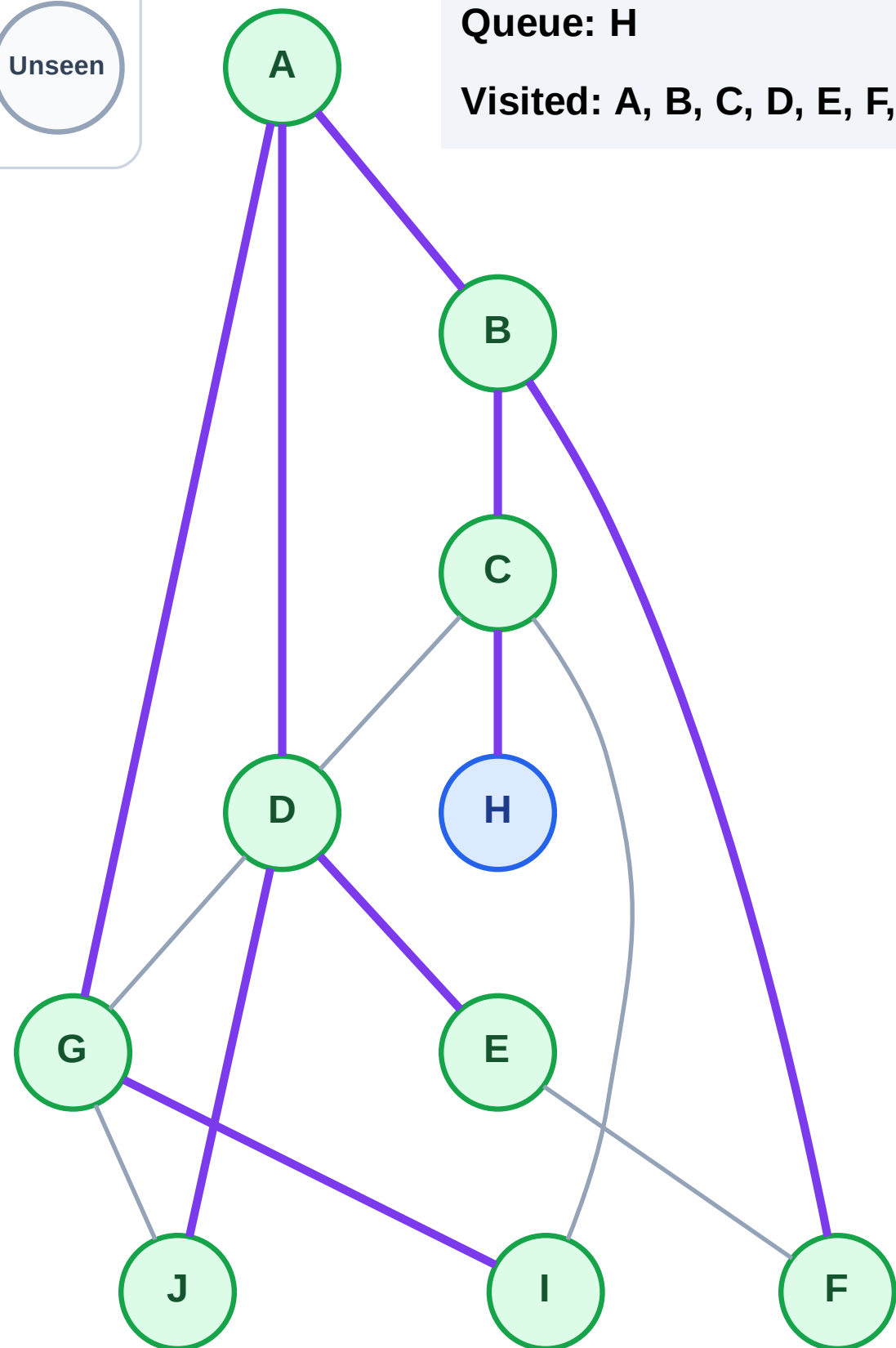
Step 19: No new neighbors from I

Legend



Queue: H

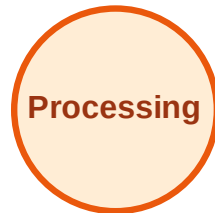
Visited: A, B, C, D, E, F, G, I, J



BFS Traversal

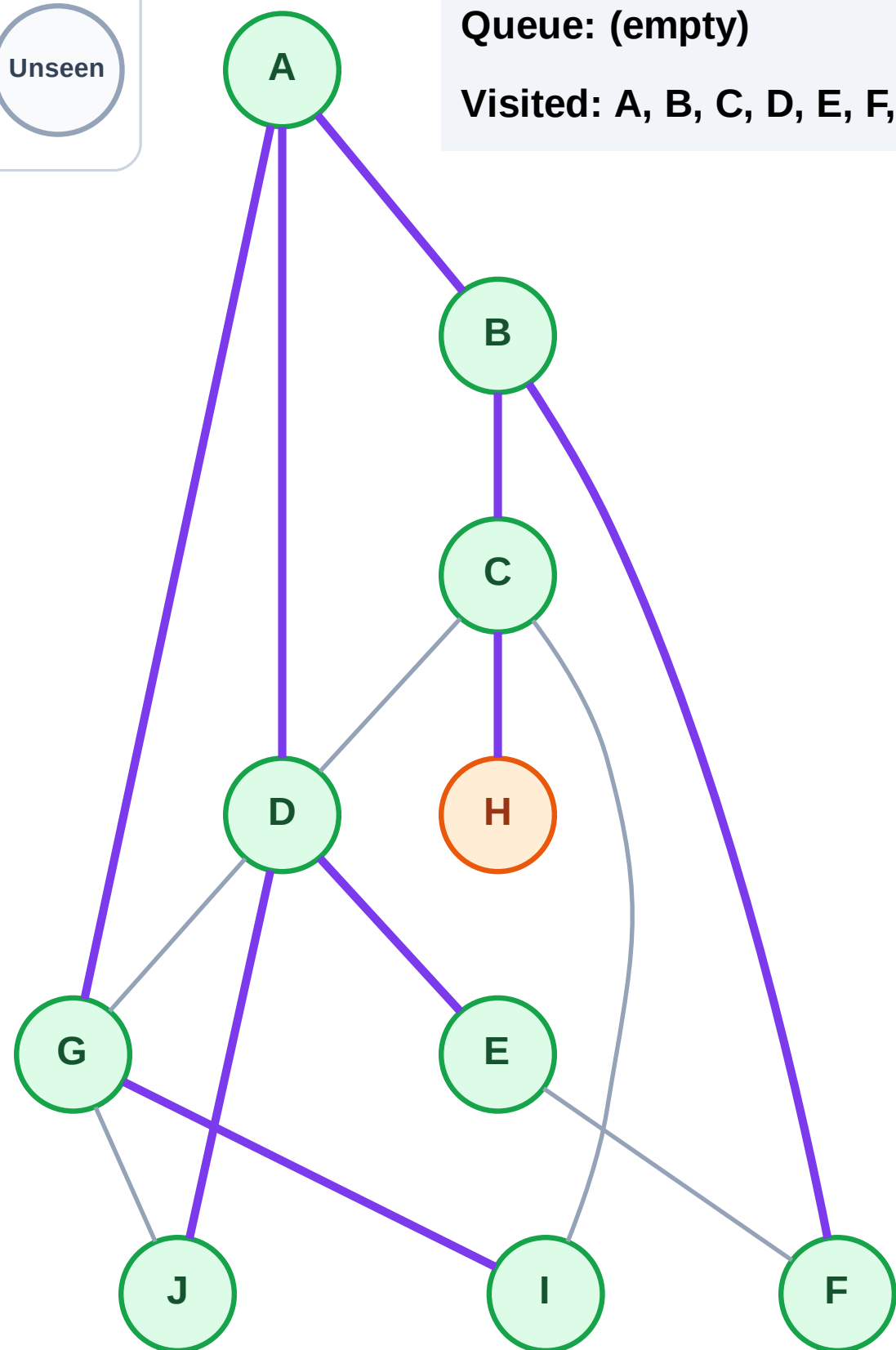
Step 20: Process node H

Legend



Queue: (empty)

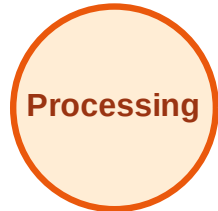
Visited: A, B, C, D, E, F, G, I, J



BFS Traversal

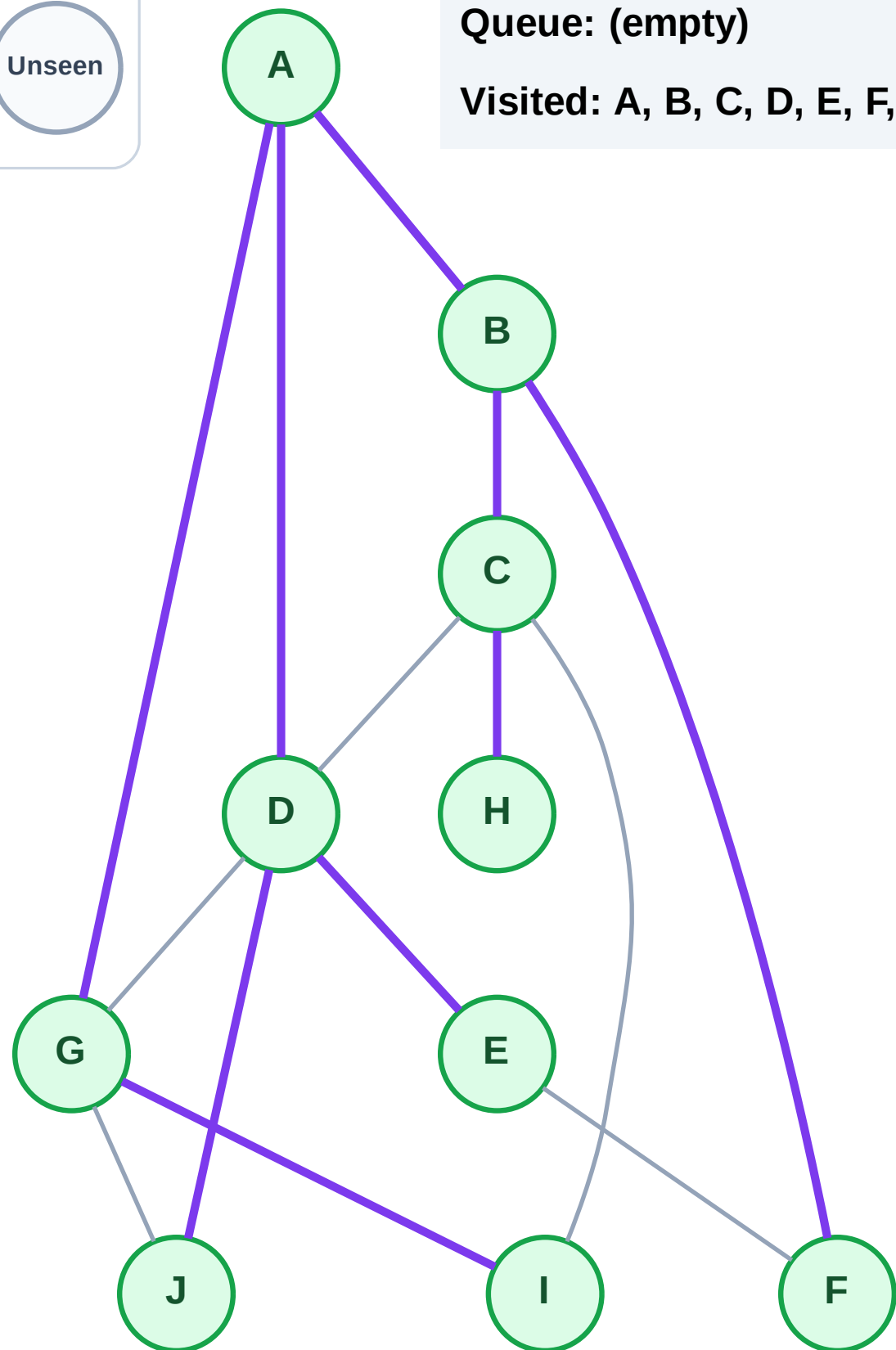
Step 21: No new neighbors from H

Legend



Queue: (empty)

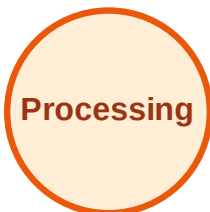
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

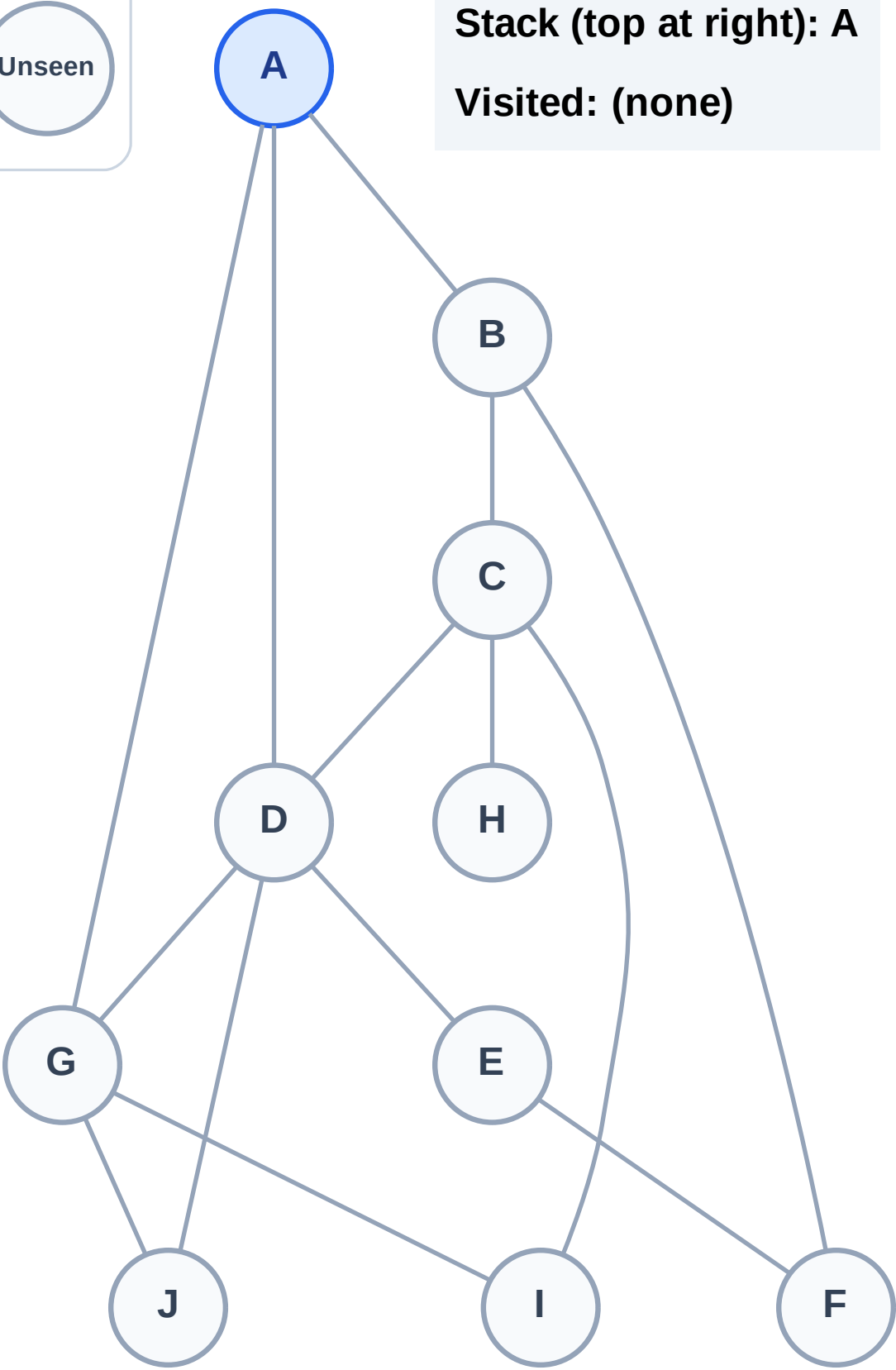
Step 1: Start at A

Legend



Stack (top at right): A

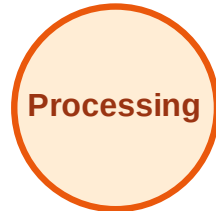
Visited: (none)



DFS Traversal

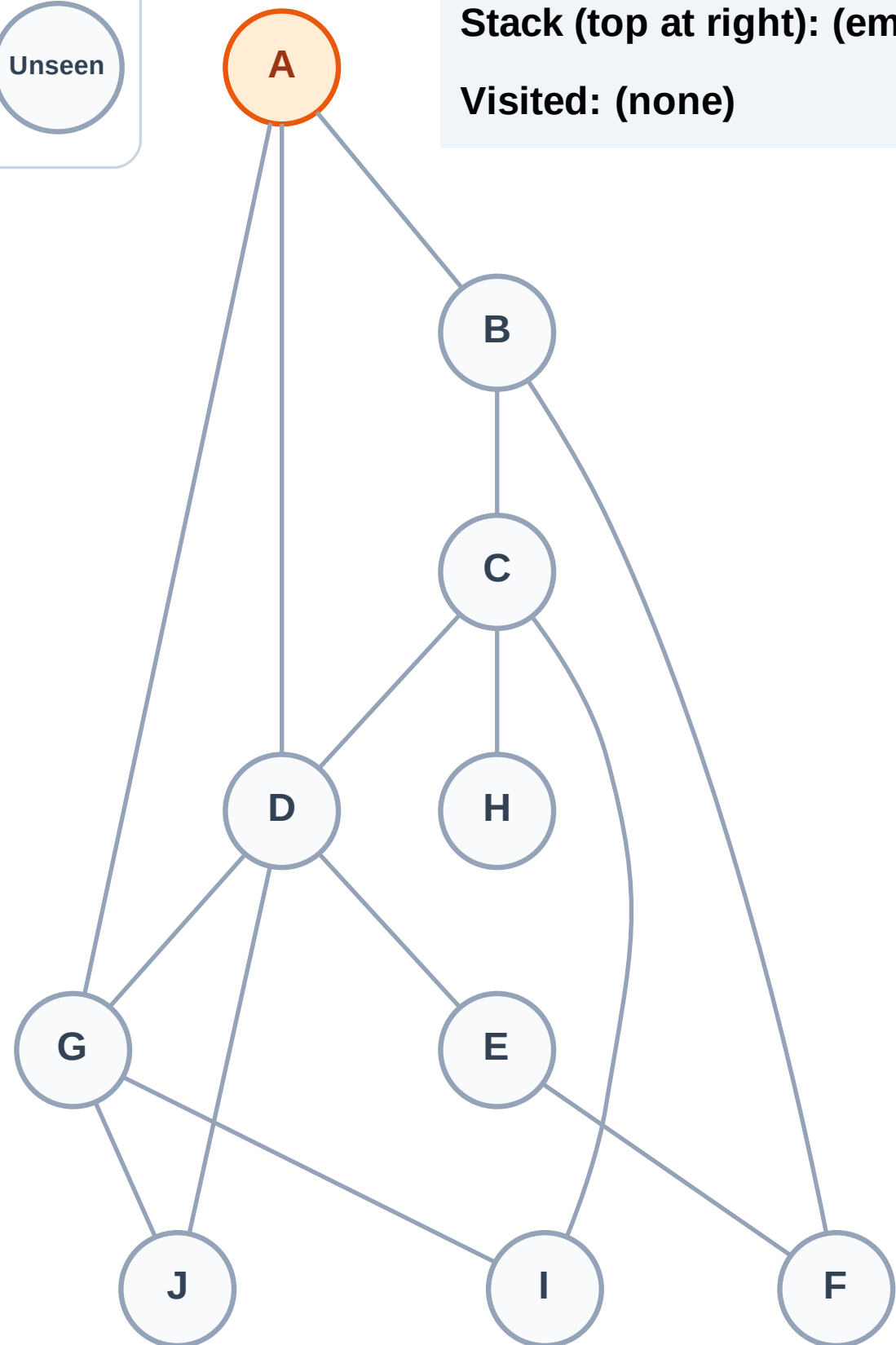
Step 2: Process node A

Legend



Stack (top at right): (empty)

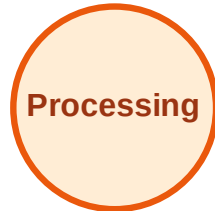
Visited: (none)



DFS Traversal

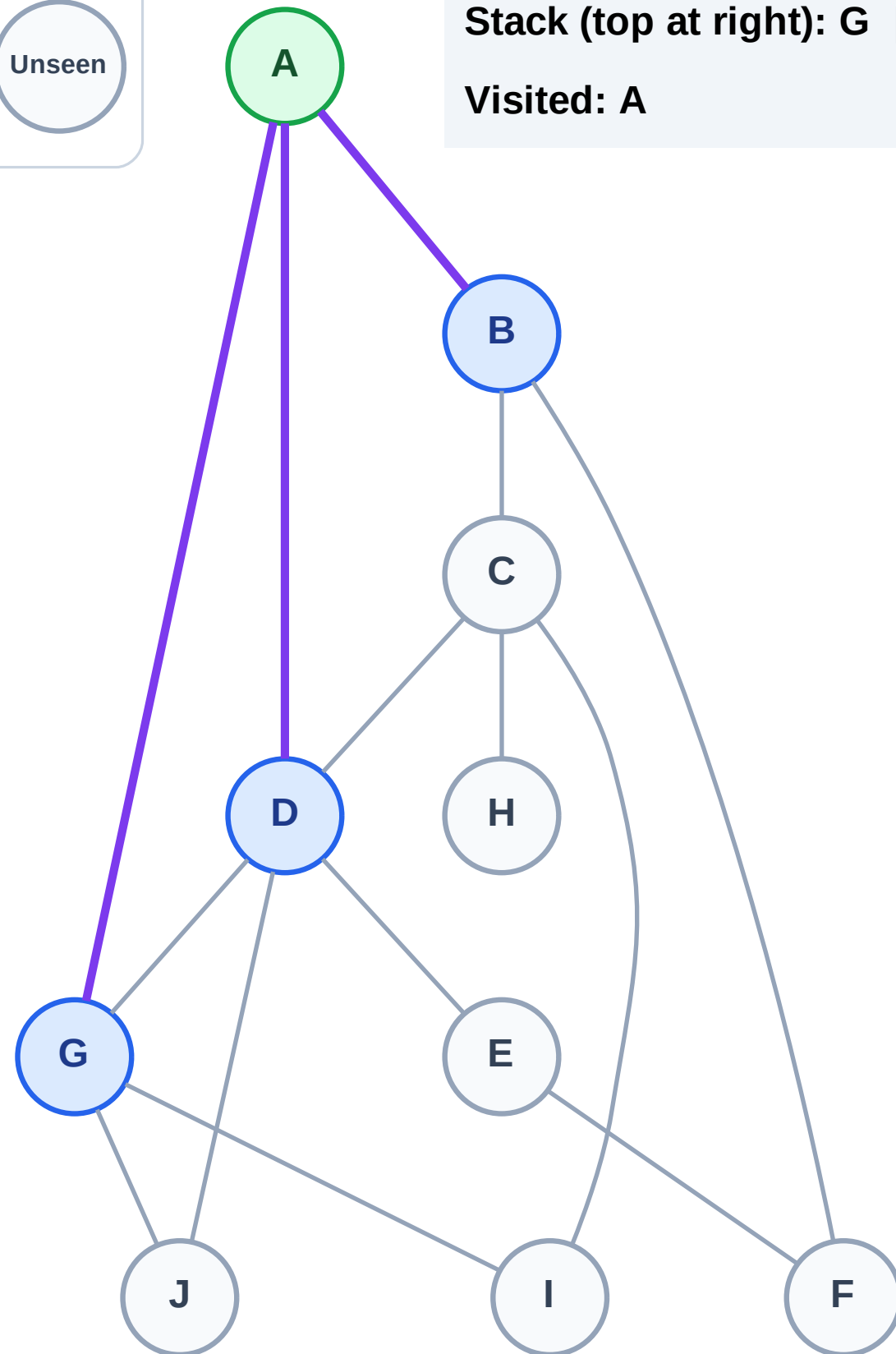
Step 3: Discover G, D, B from A

Legend



Stack (top at right): G | D | B

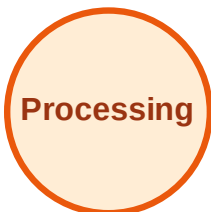
Visited: A



DFS Traversal

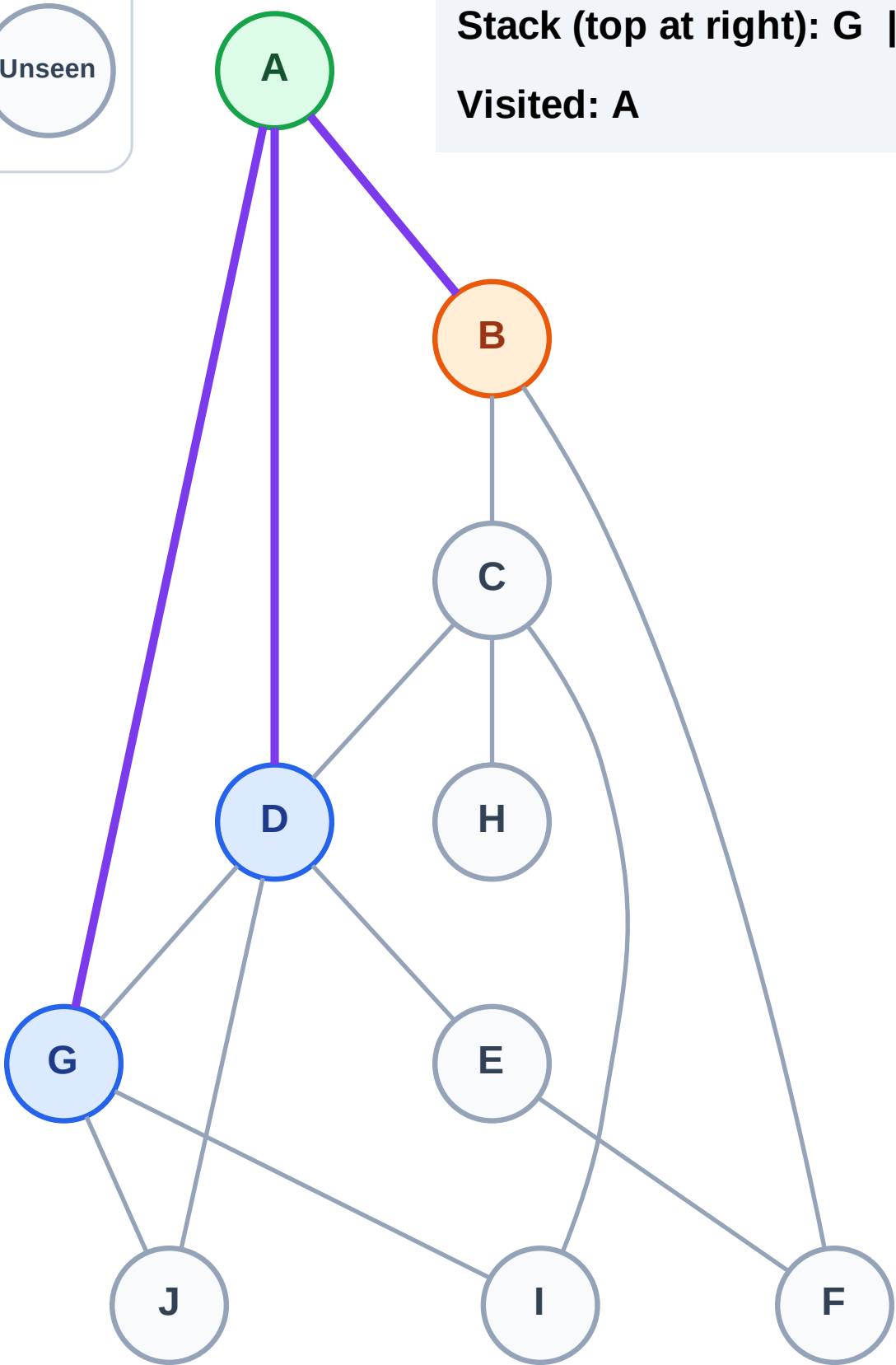
Step 4: Process node B

Legend



Stack (top at right): G | D

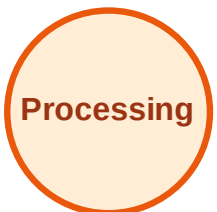
Visited: A



DFS Traversal

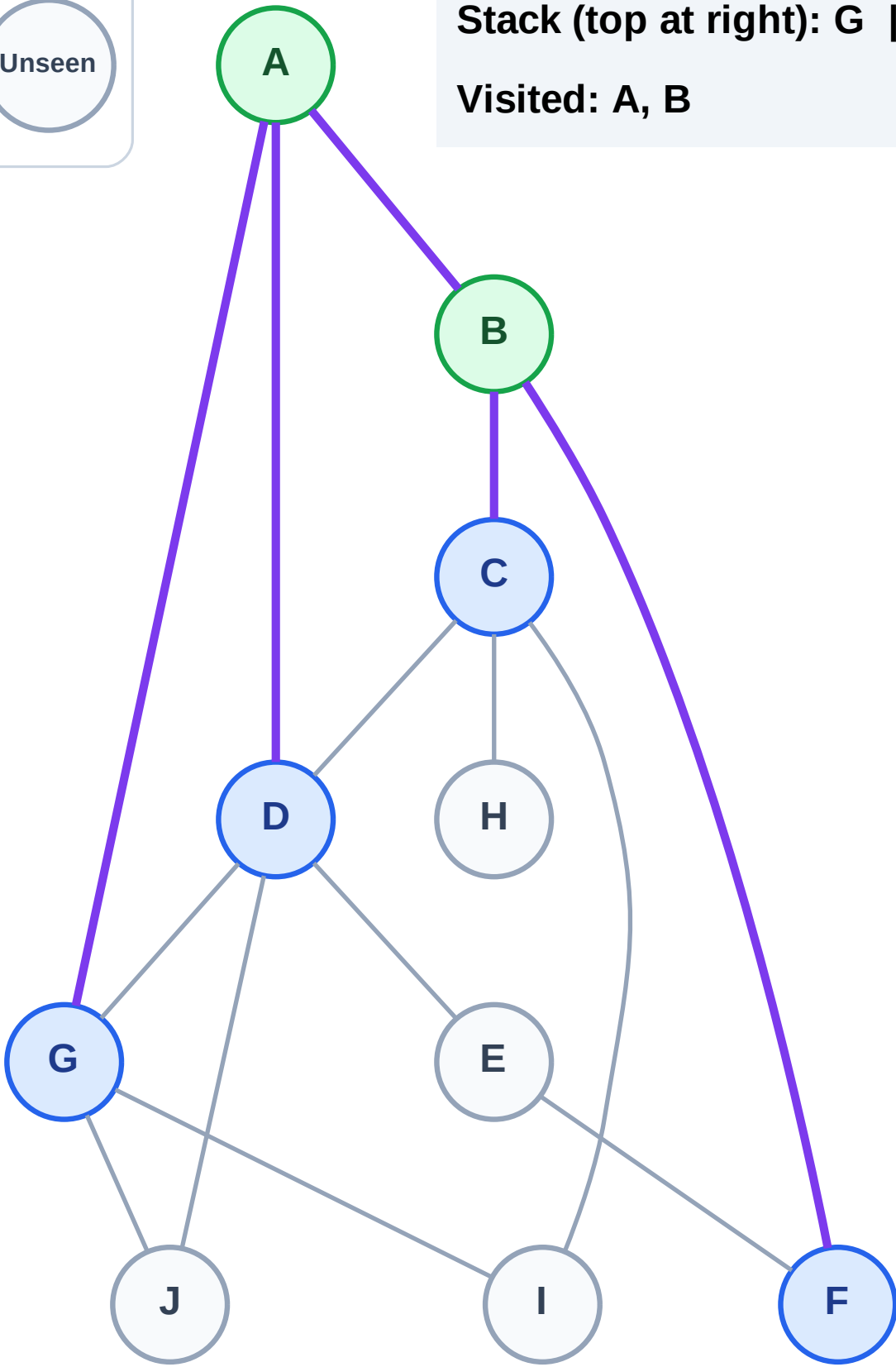
Step 5: Discover F, C from B

Legend



Stack (top at right): G | D | F | C

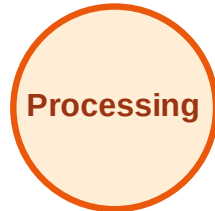
Visited: A, B



DFS Traversal

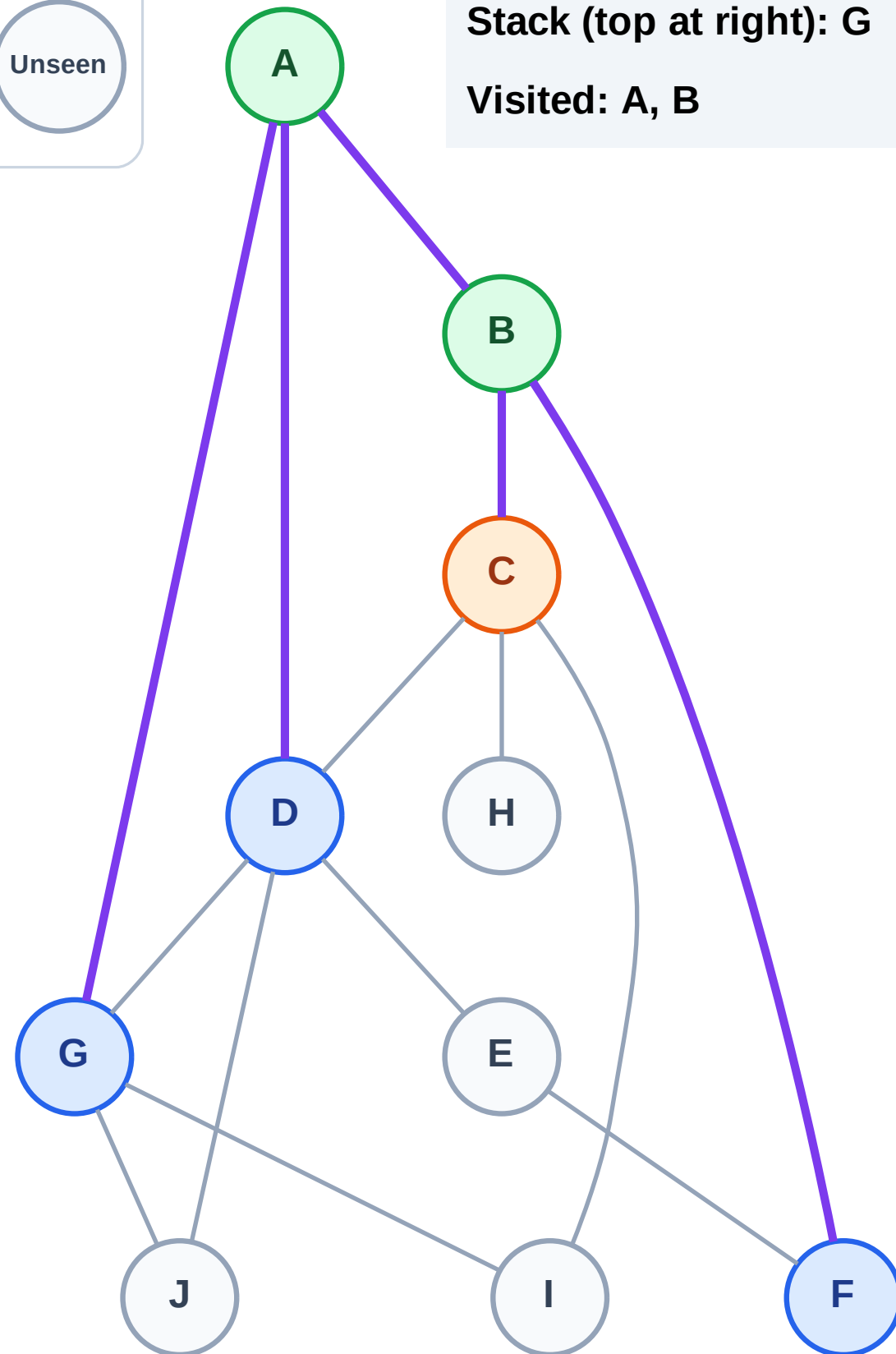
Step 6: Process node C

Legend



Stack (top at right): G | D | F

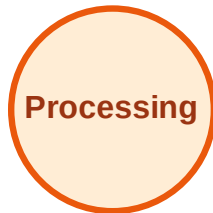
Visited: A, B



DFS Traversal

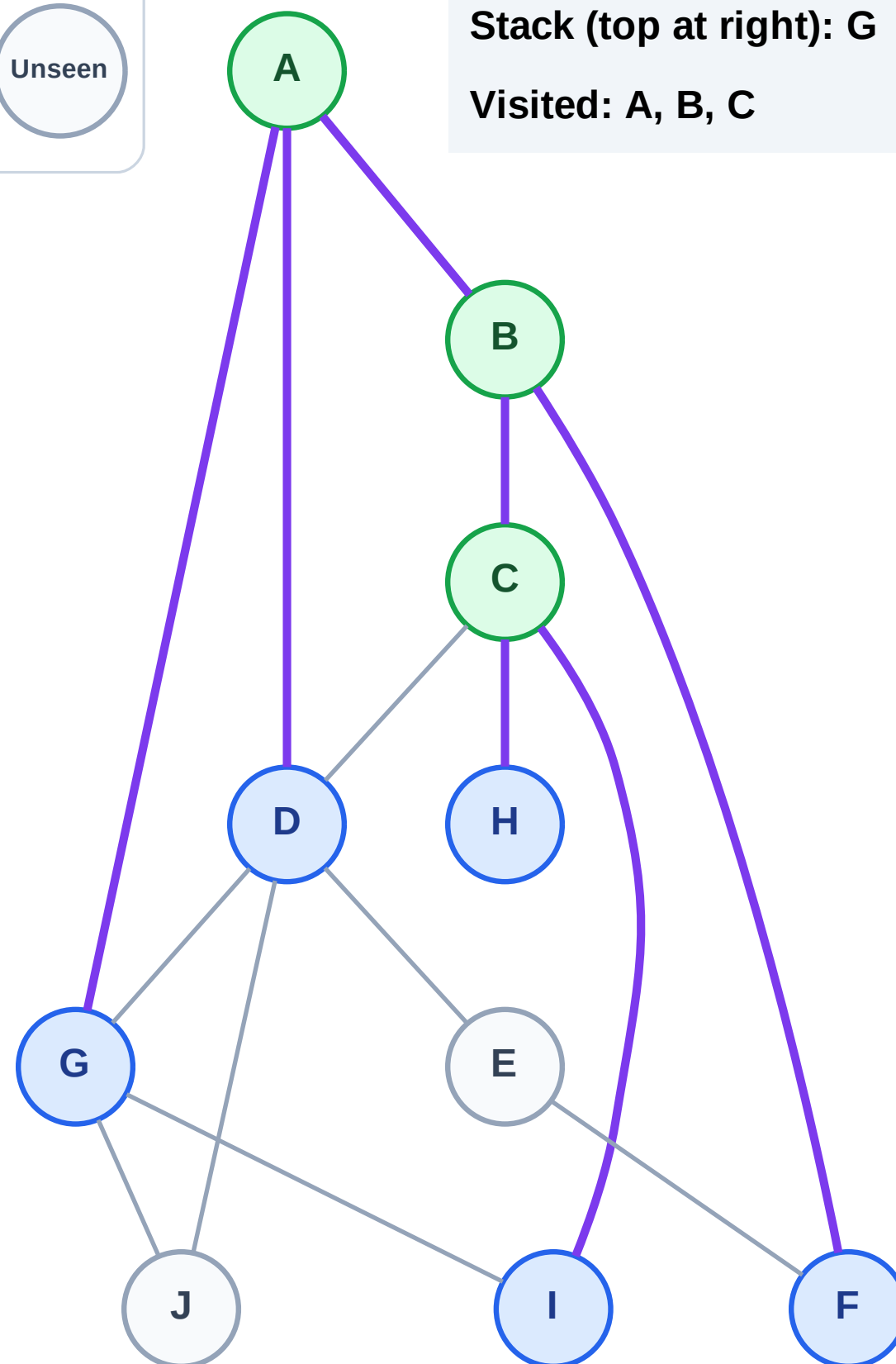
Step 7: Discover I, H from C

Legend



Stack (top at right): G | D | F | I | H

Visited: A, B, C



DFS Traversal

Step 8: Process node H

Legend

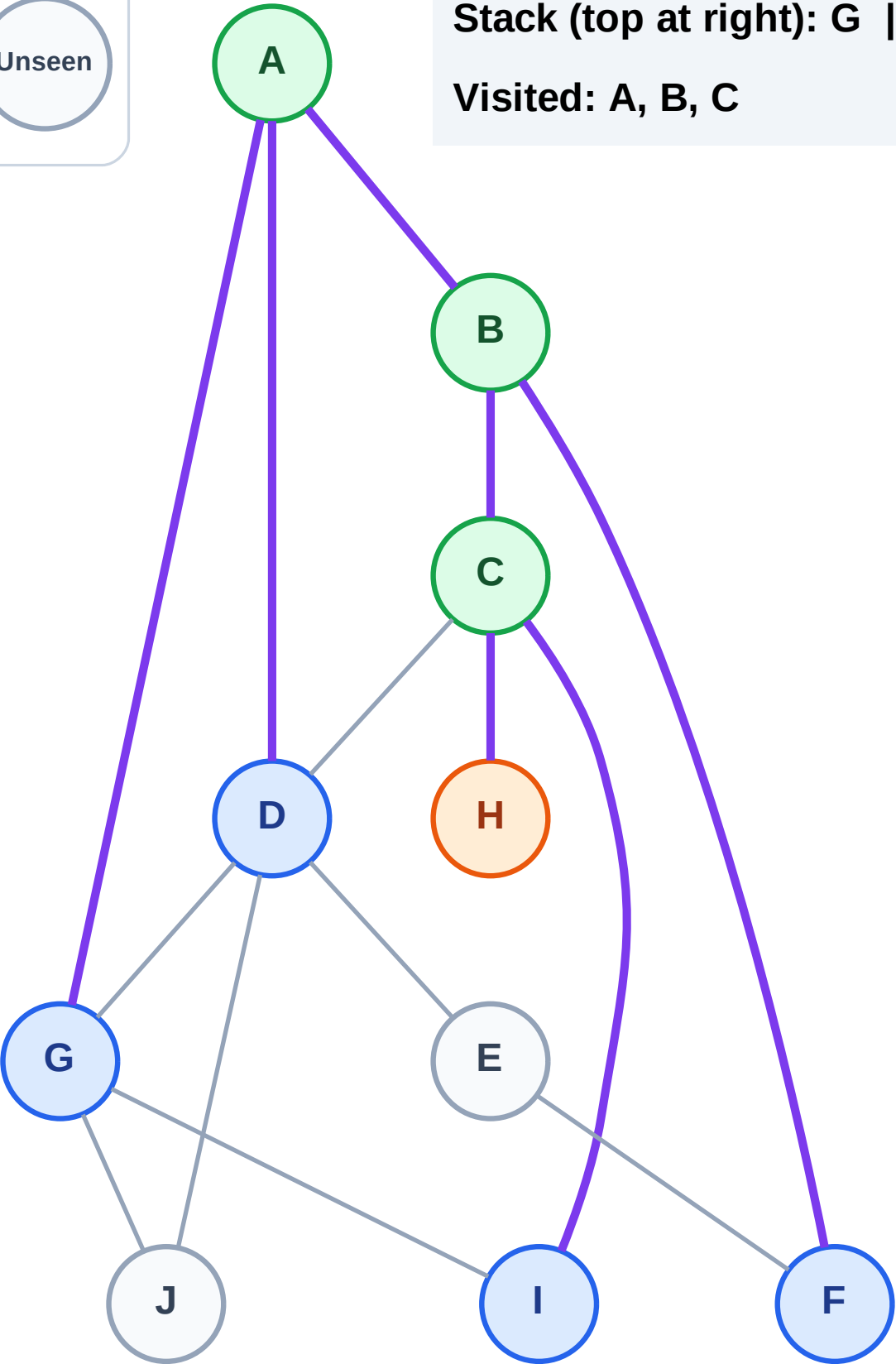
Complete

Processing

Discovered

Unseen

Stack (top at right): G | D | F | I
Visited: A, B, C



DFS Traversal

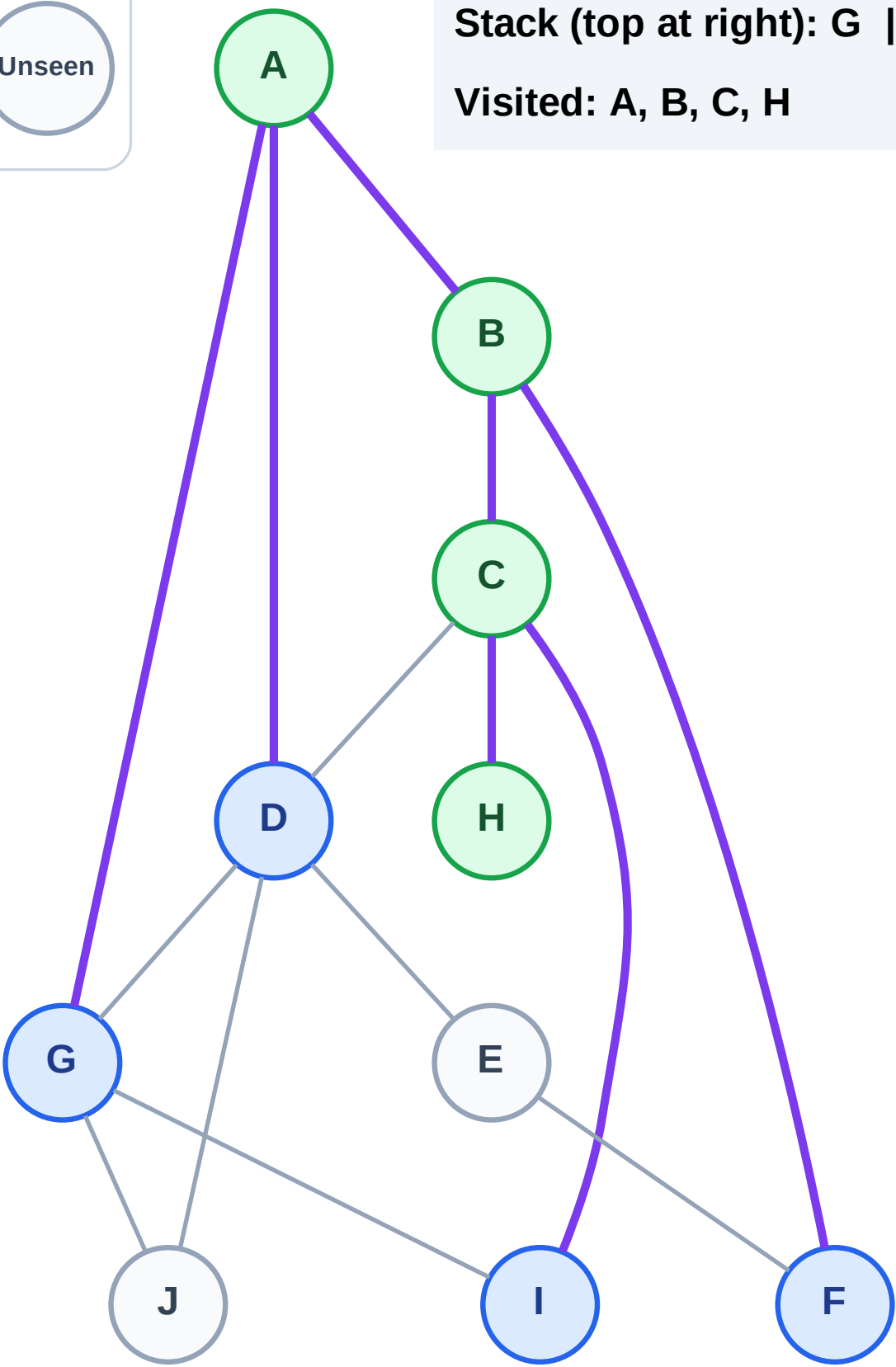
Step 9: No new neighbors from H

Legend



Stack (top at right): G | D | F | I

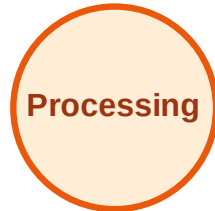
Visited: A, B, C, H



DFS Traversal

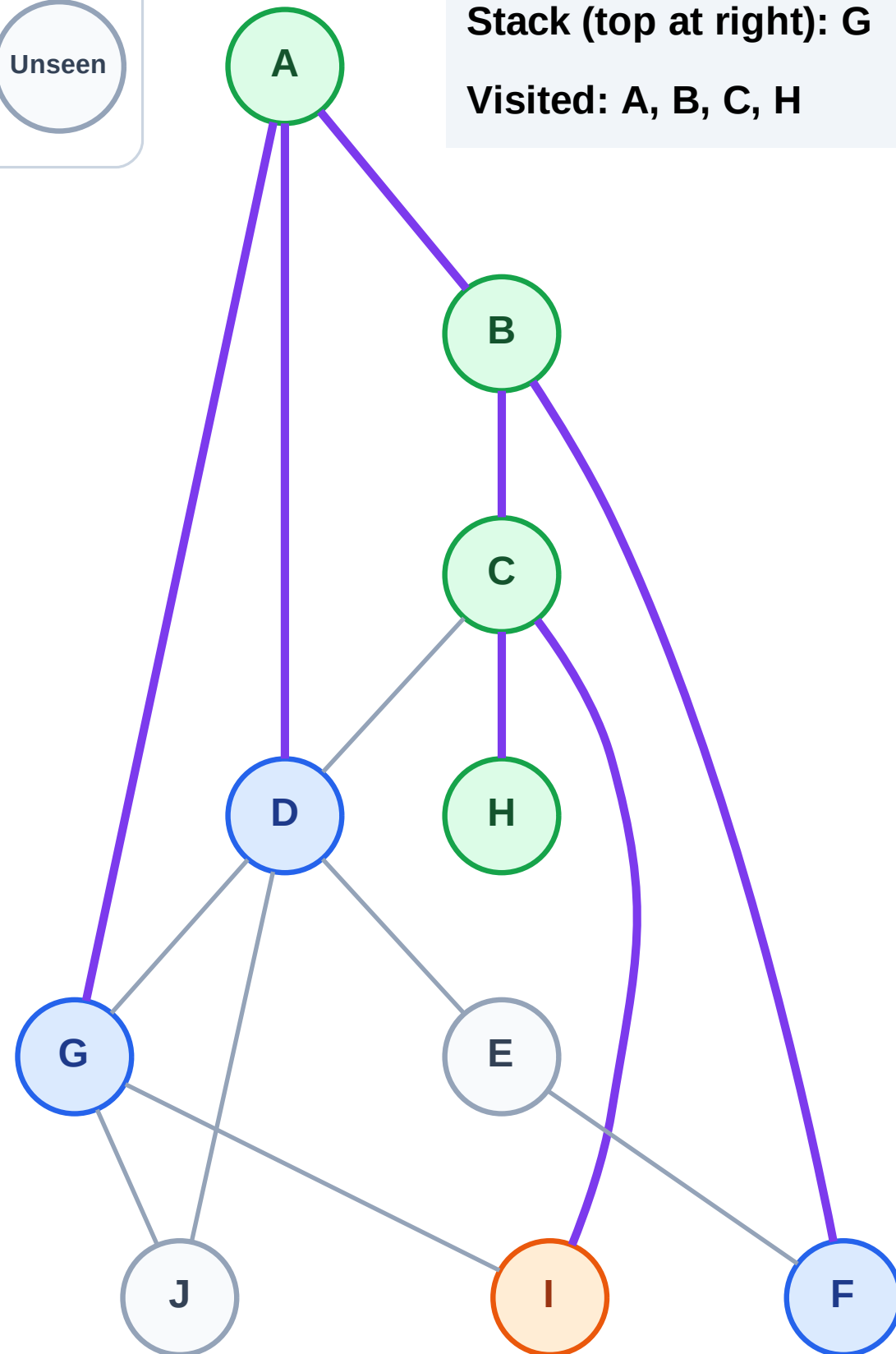
Step 10: Process node I

Legend



Stack (top at right): G | D | F

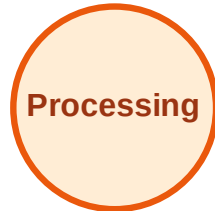
Visited: A, B, C, H



DFS Traversal

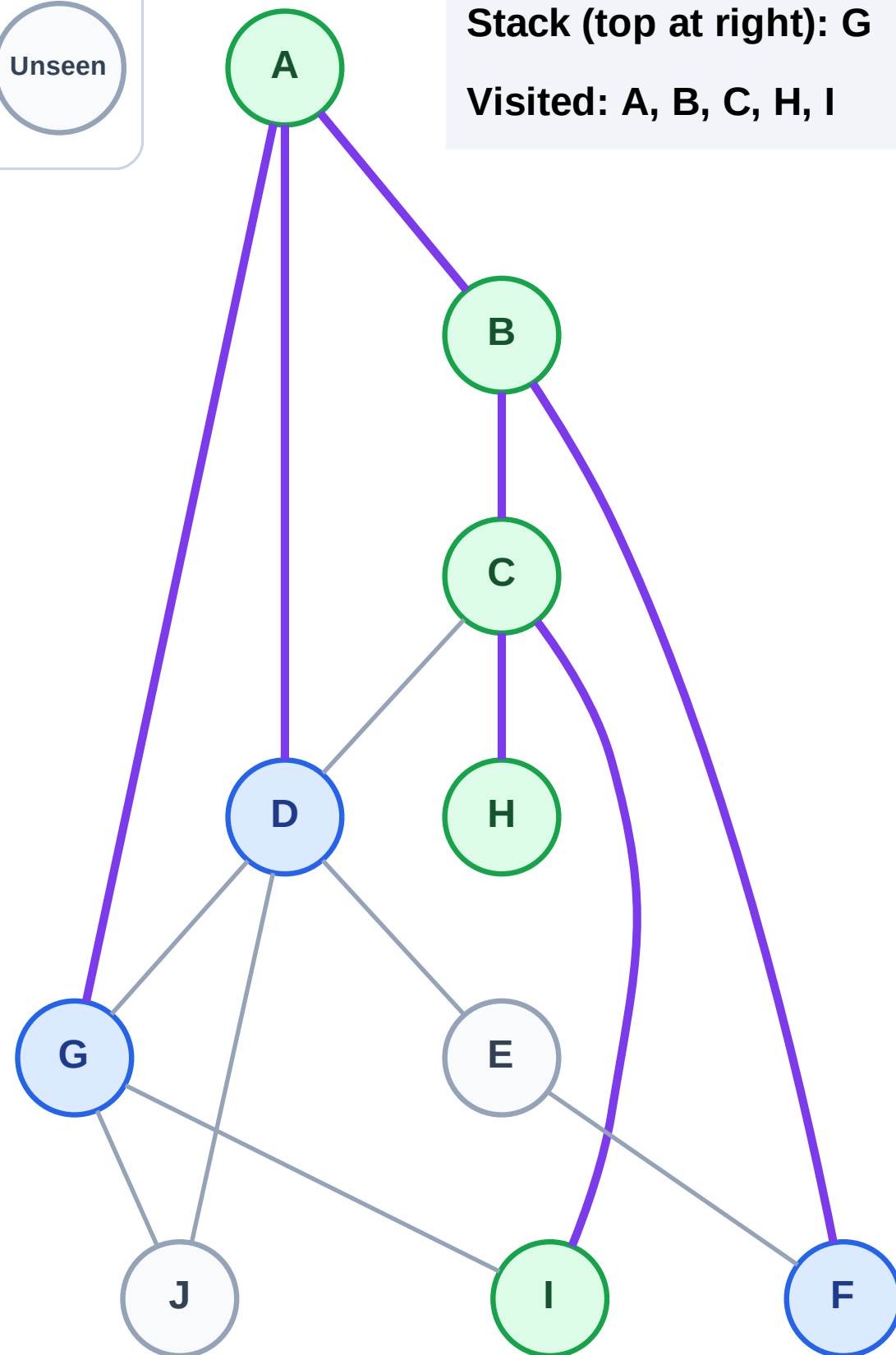
Step 11: No new neighbors from I

Legend



Stack (top at right): G | D | F

Visited: A, B, C, H, I



DFS Traversal

Step 12: Process node F

Legend

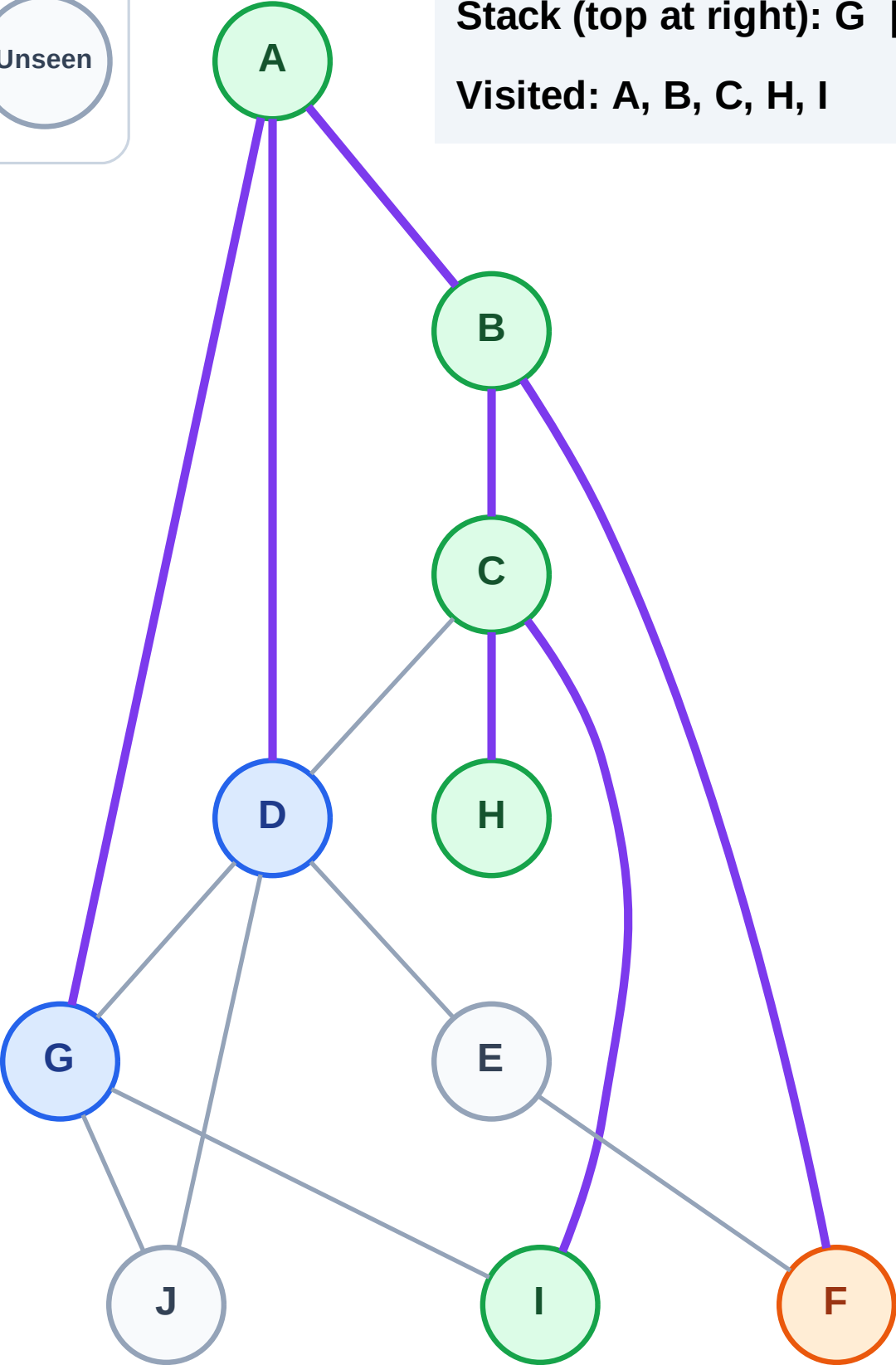
Complete

Processing

Discovered

Unseen

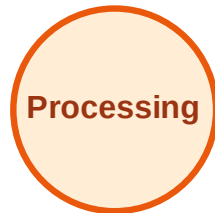
Stack (top at right): G | D
Visited: A, B, C, H, I



DFS Traversal

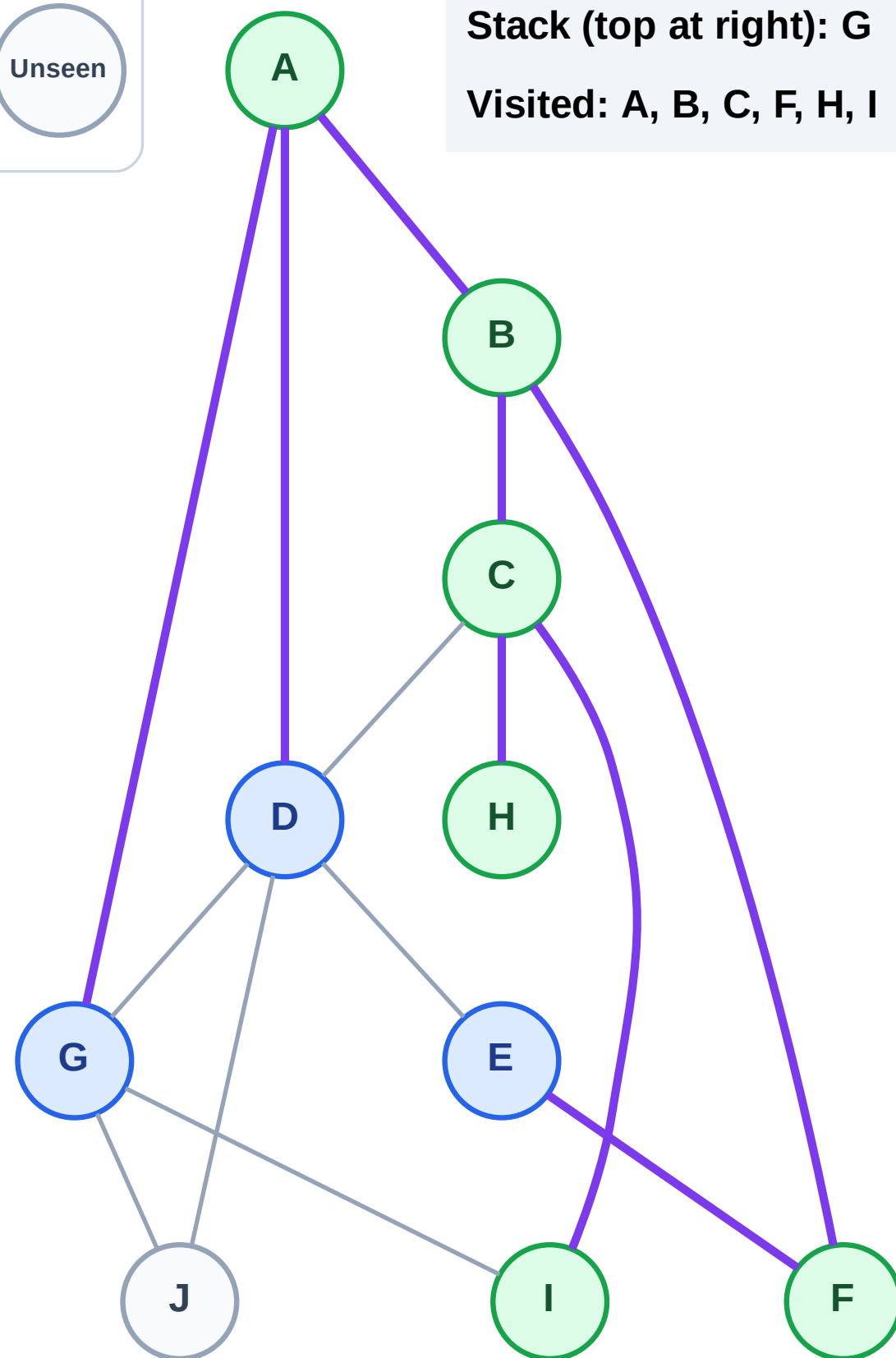
Step 13: Discover E from F

Legend



Stack (top at right): G | D | E

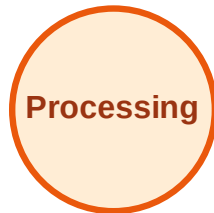
Visited: A, B, C, F, H, I



DFS Traversal

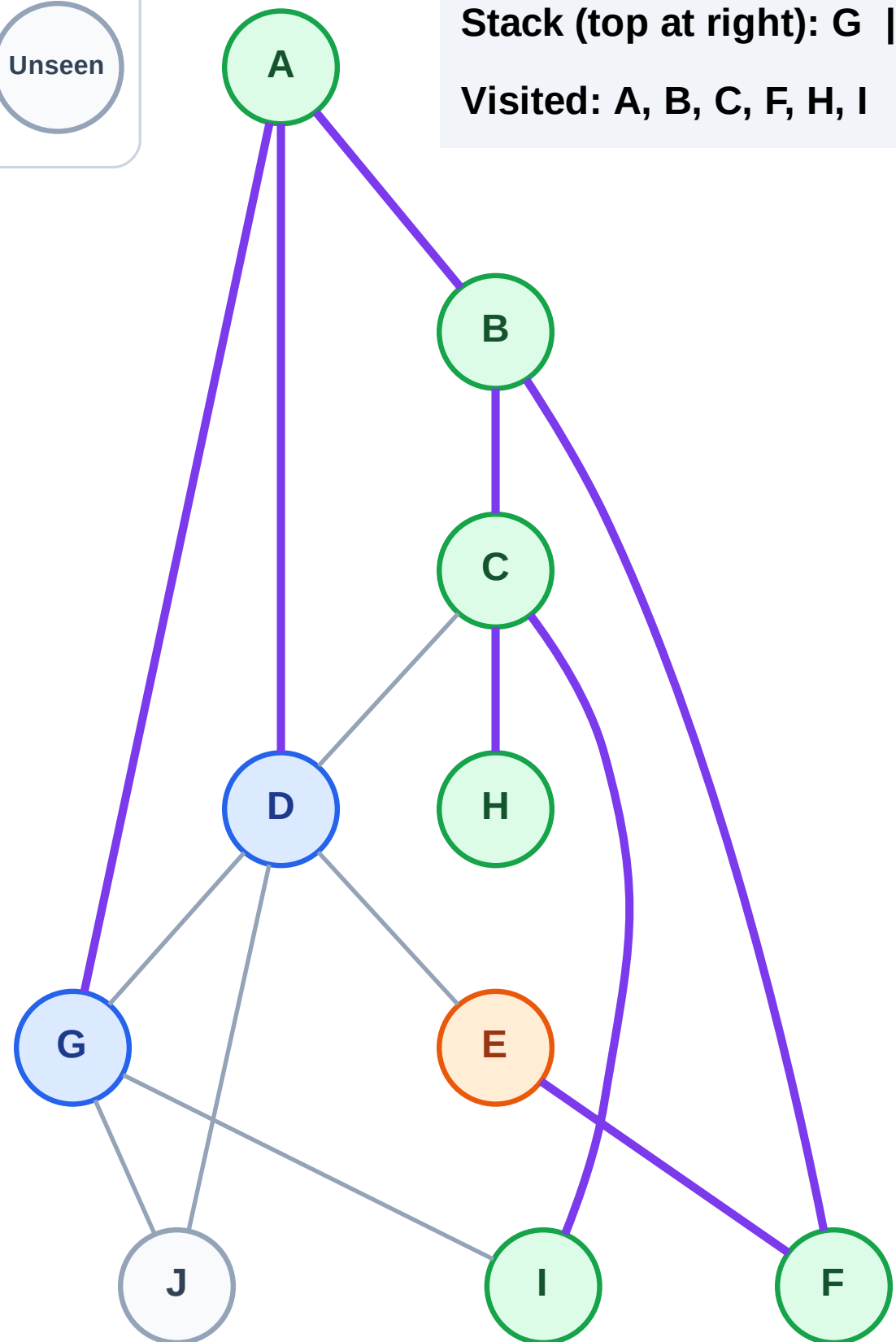
Step 14: Process node E

Legend



Stack (top at right): G | D

Visited: A, B, C, F, H, I



DFS Traversal

Step 15: No new neighbors from E

Legend

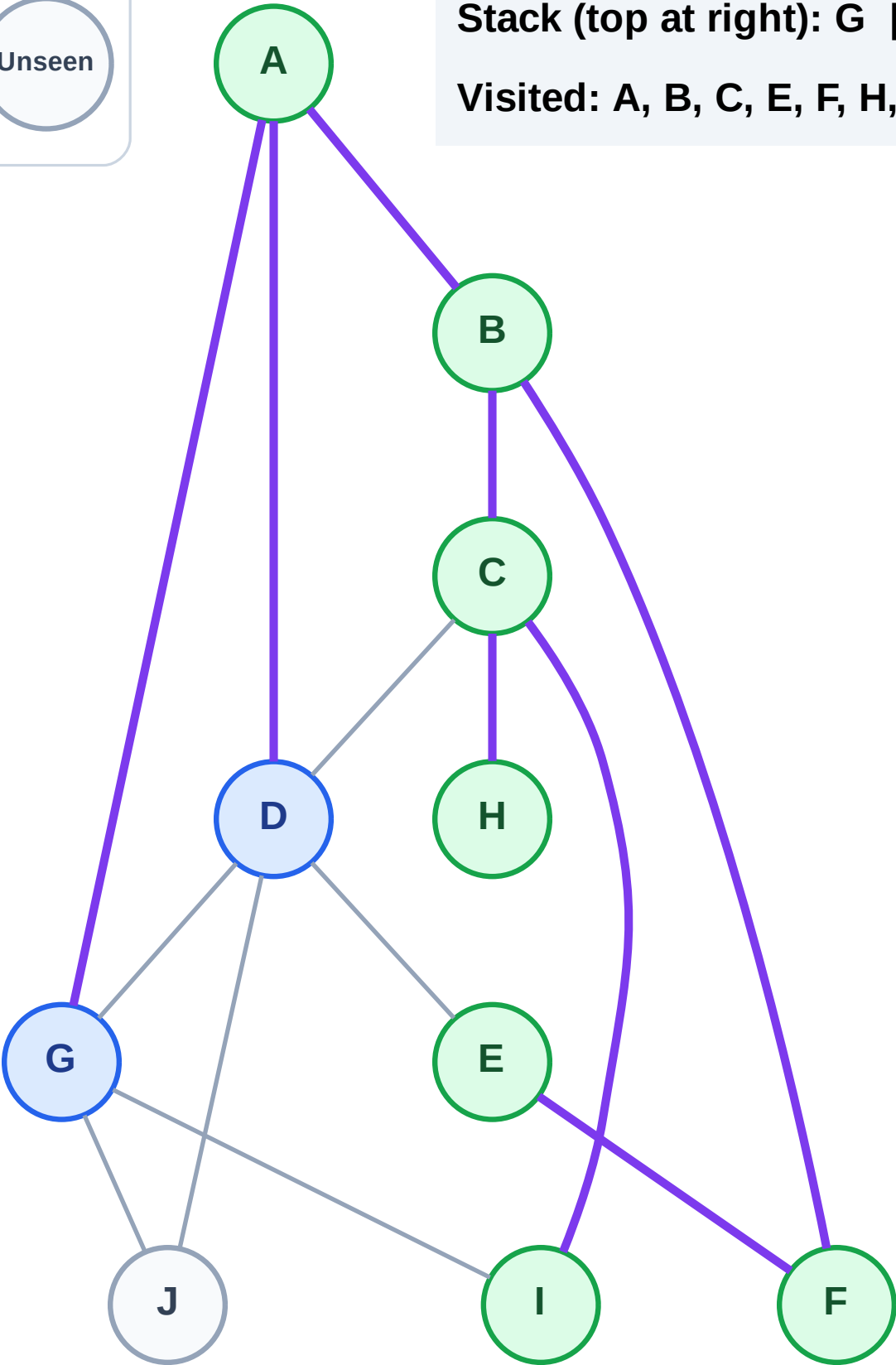
Complete

Processing

Discovered

Unseen

Stack (top at right): G | D
Visited: A, B, C, E, F, H, I



DFS Traversal

Step 16: Process node D

Legend

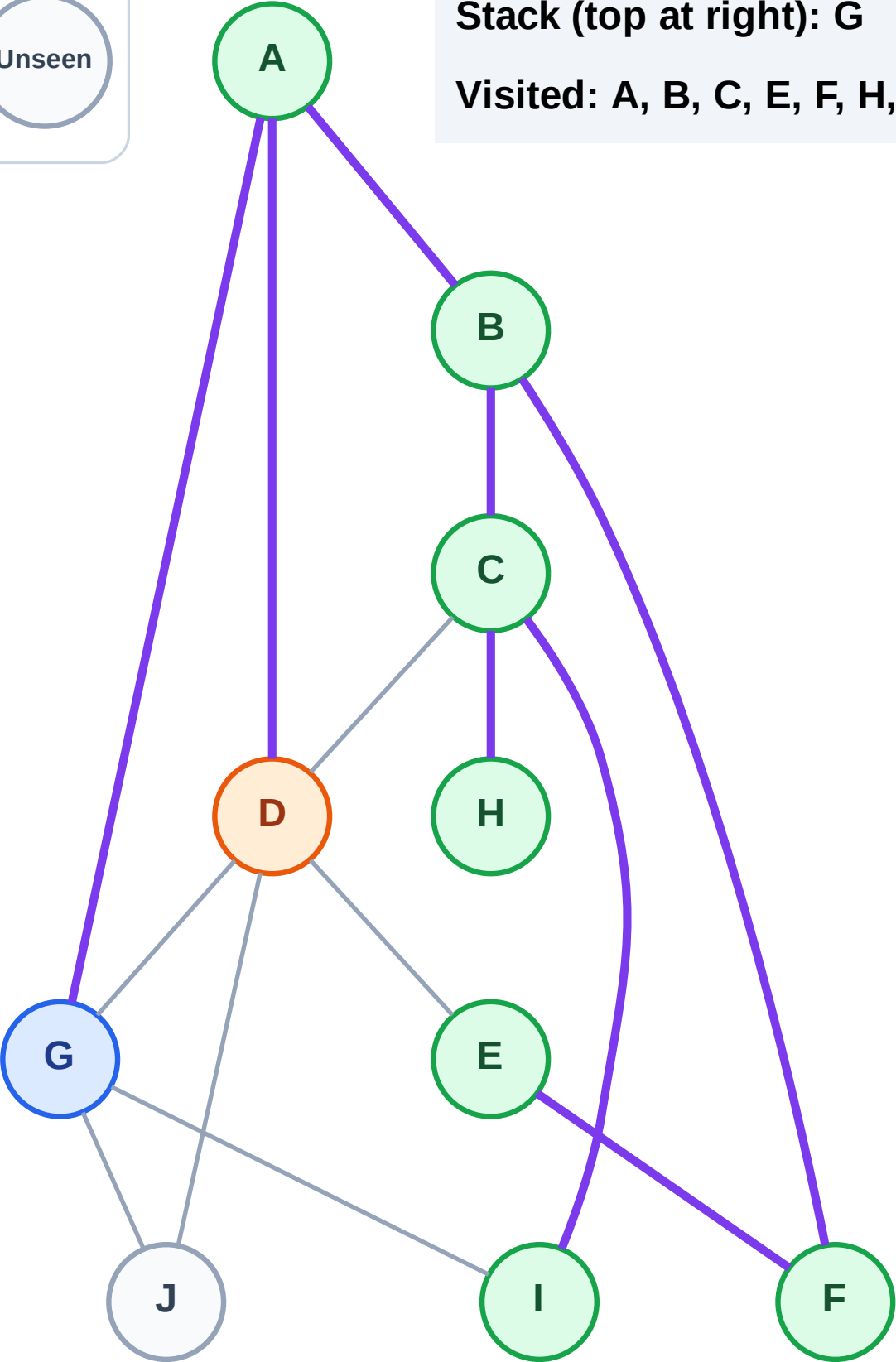
Complete

Processing

Discovered

Unseen

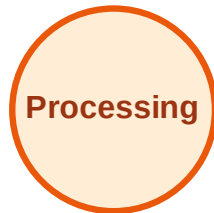
Stack (top at right): G
Visited: A, B, C, E, F, H, I



DFS Traversal

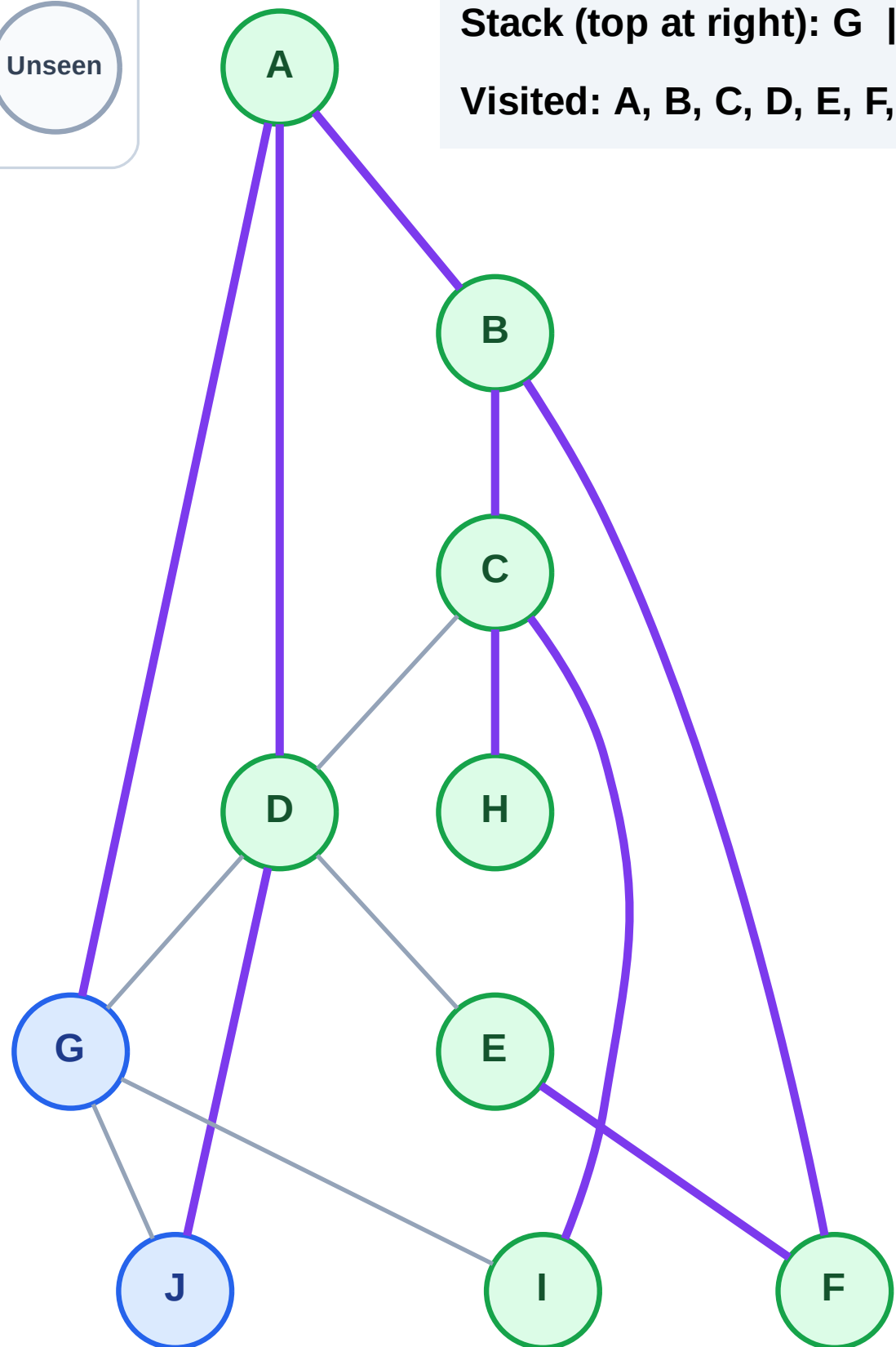
Step 17: Discover J from D

Legend



Stack (top at right): G | J

Visited: A, B, C, D, E, F, H, I



DFS Traversal

Step 18: Process node J

Legend

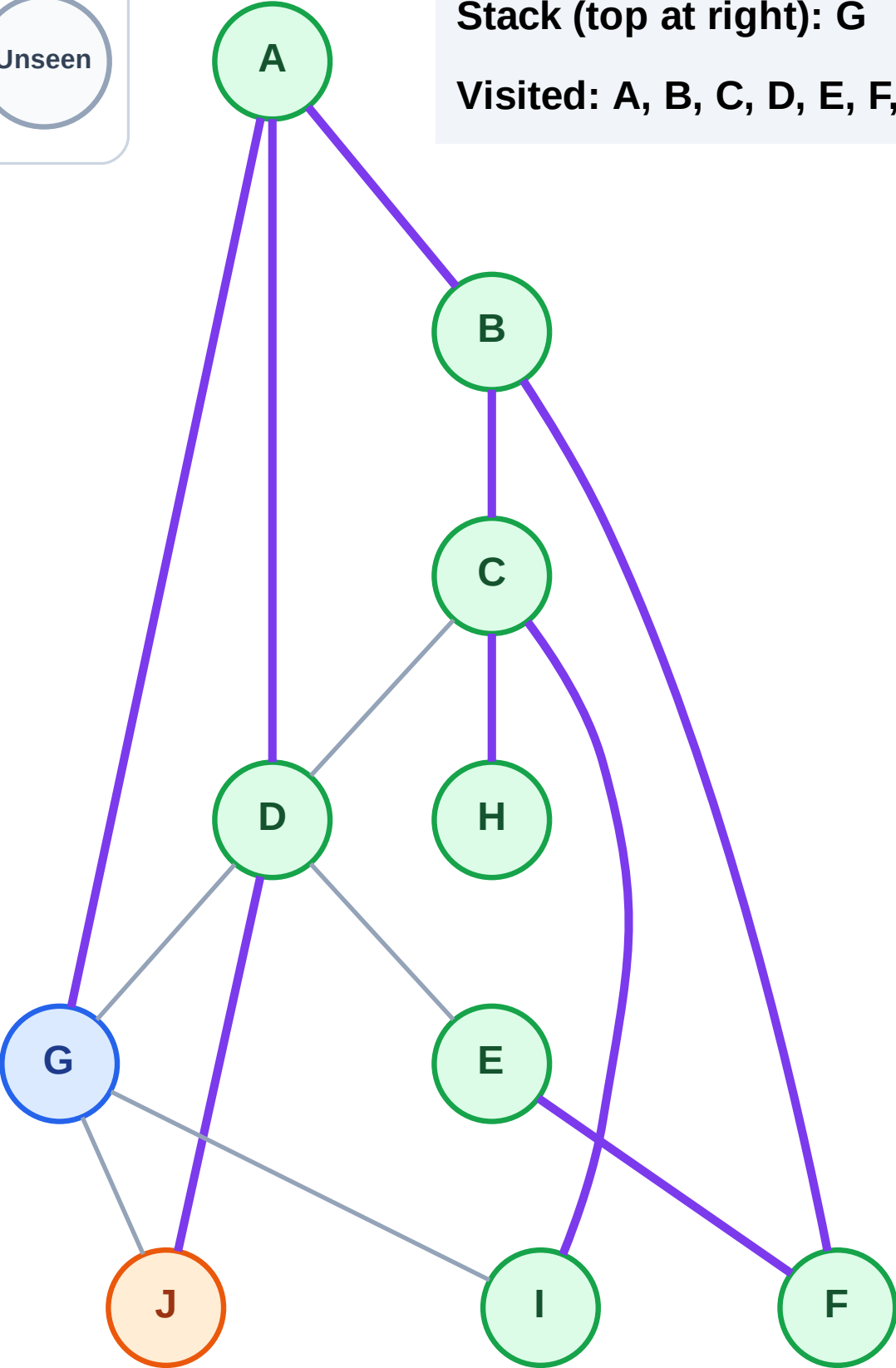
Complete

Processing

Discovered

Unseen

Stack (top at right): G
Visited: A, B, C, D, E, F, H, I

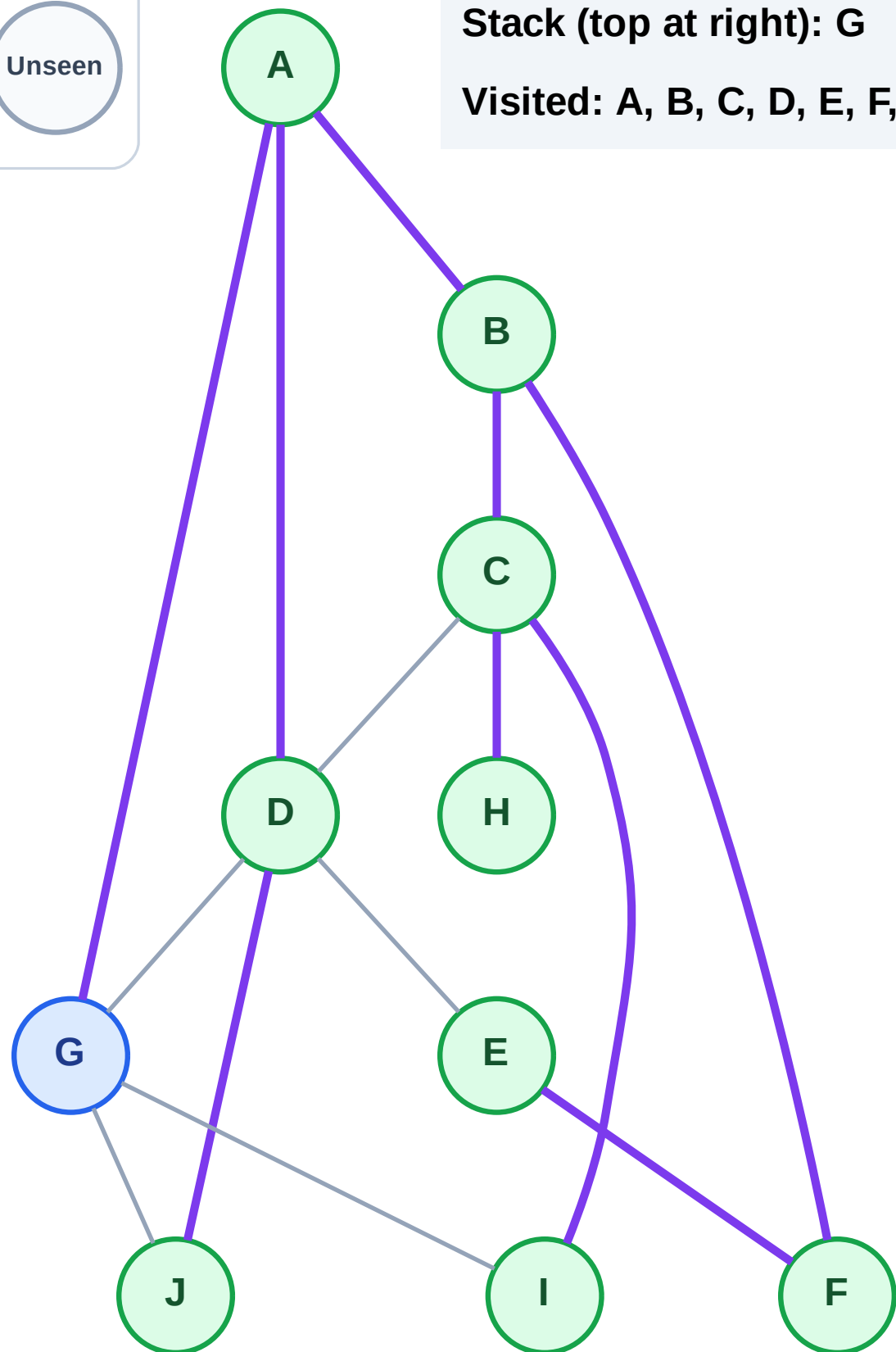


Step 19: No new neighbors from J

Legend

Unseen

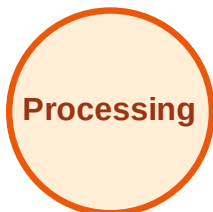
Visited: A, B, C, D, E, F, H, I, J



DFS Traversal

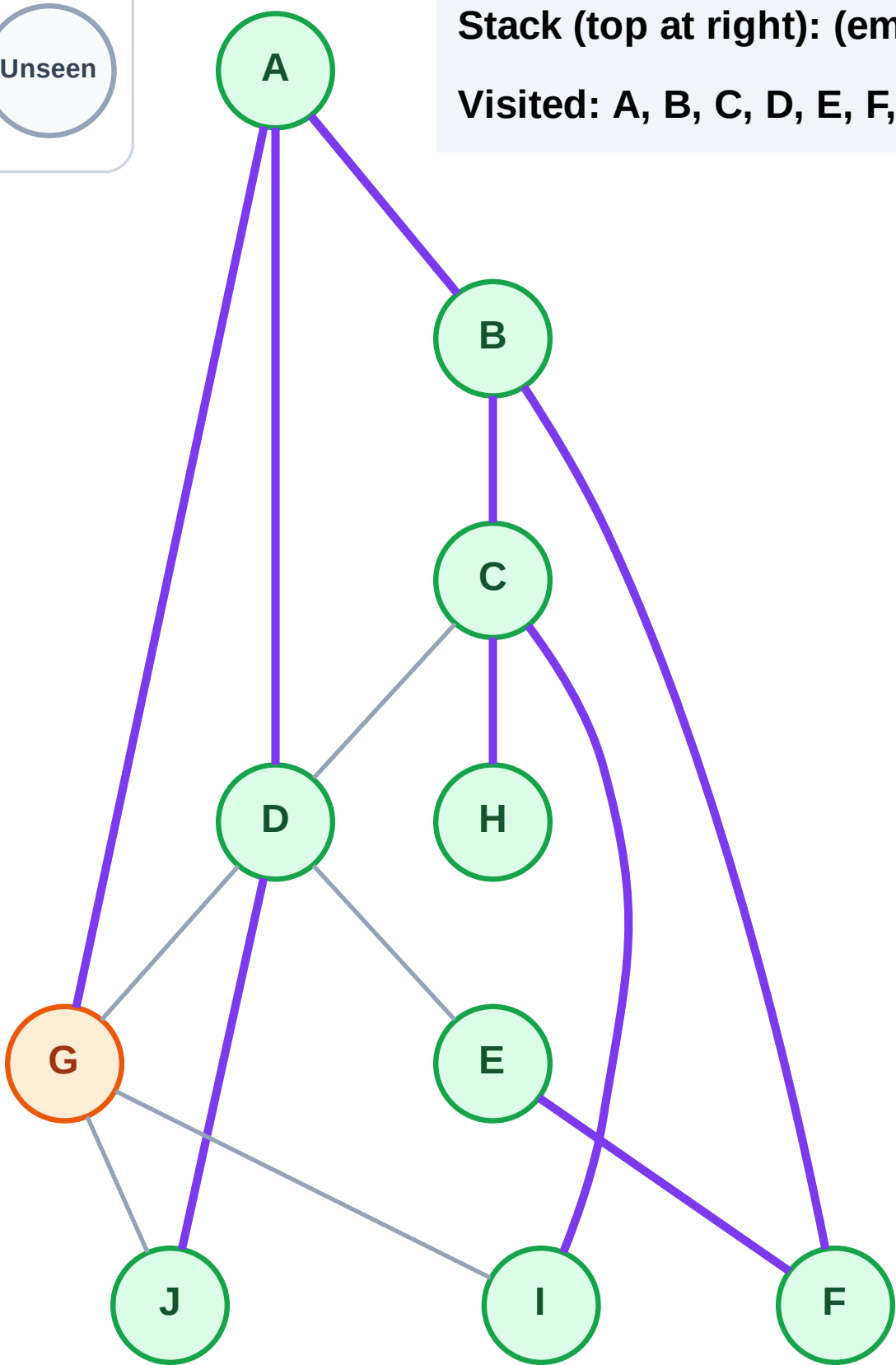
Step 20: Process node G

Legend



Stack (top at right): (empty)

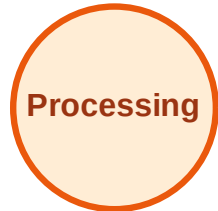
Visited: A, B, C, D, E, F, H, I, J



DFS Traversal

Step 21: No new neighbors from G

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

